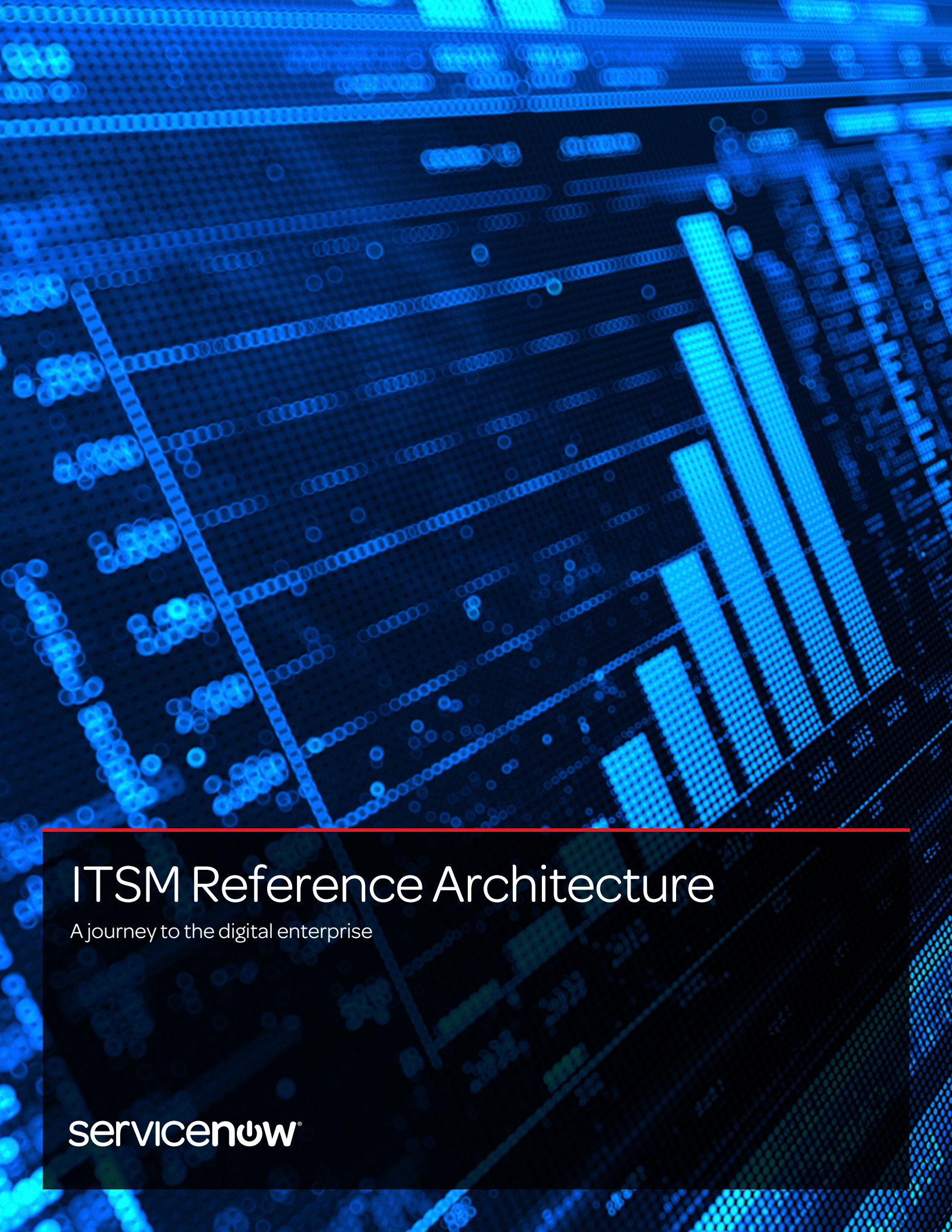


The background is a dark blue field filled with glowing digital patterns. On the left, there are vertical and diagonal lines of small circles, resembling data streams or network connections. On the right, a 3D bar chart with five bars of increasing height is visible, also composed of glowing dots. The overall aesthetic is high-tech and data-driven.

ITSM Reference Architecture

A journey to the digital enterprise

servicenow

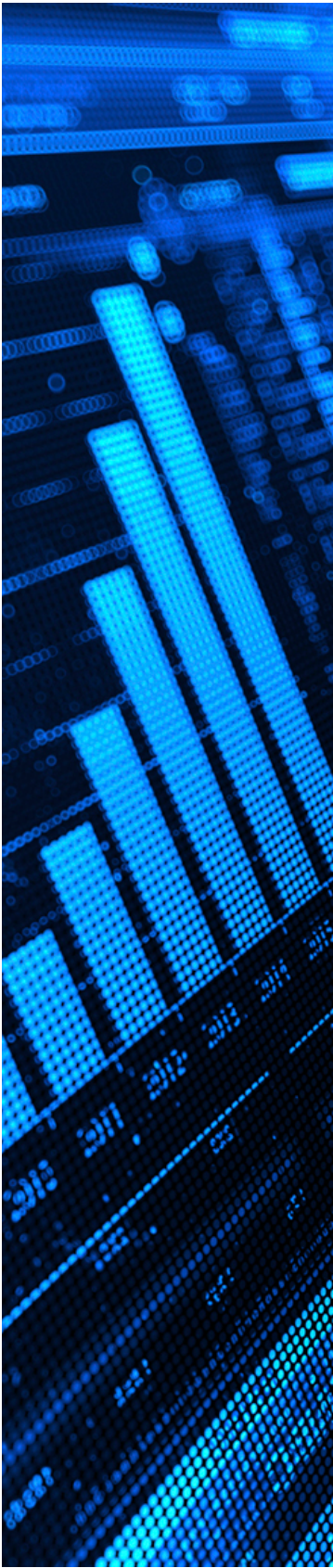


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As a customer or partner of ServiceNow, you are encouraged to combine elements of the various framework methodologies with the ServiceNow technology stack for strategic advantage.

Purpose and Objectives

With thousands of data collection points spanning every market, customers have asked ServiceNow® to share key learnings and best practice guidance to help illuminate the path to value realization, accelerate time to value, and to help them realize maximum return on investment (ROI).

The intended outcome of this eBook is to take customers through a series of planning activities that are designed to flesh out the plans for a successful IT service management (ITSM) program that is tailored to your organization and harmonized within the ServiceNow technology stack. This eBook will be decoupled into more targeted conversations and released via ServiceNow communities and the Customer Success Center as well.

As a customer or partner of ServiceNow, you are encouraged to combine elements of the various framework methodologies with the ServiceNow technology stack for strategic advantage. One of the core tenets of the ITSM Reference Architecture is to serve as a harmonizing force by bridging the white space between theory and practice.

The ServiceNow guidance is to blend elements of the following frameworks including the architecture review board (ARB) and architectural principles of TOGAF, the service management principles of ITIL, the service backbone and common data model from the IT4IT standard, the digital perspectives of VeriSM, the agile concepts from the Scaled Agile Framework (SAFe), and the culture that thrives within DevOps.

ServiceNow is uniquely positioned to help customers harmonize these framework elements to unleash the power of our modern ITSM application stack on the ServiceNow platform. The key value propositions of the ITSM Reference Architecture are **(1) accelerated time to value, (2) dramatically lower costs, and (3) maximizing the return on investment.**

ITSM reference architecture

Field-tested best practices

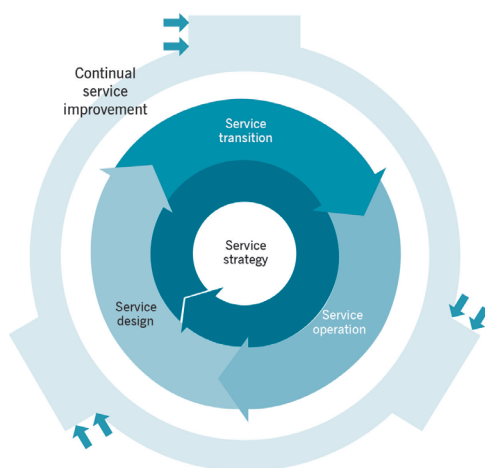
Traditionally, IT Service Management (ITSM) practitioners have been left to their devices when it comes to harmonizing industry best practice frameworks with their technology stack. By design, best practice frameworks and industry guidance tend to be technology agnostic, which introduces a layer of subjectivity when it comes to practical implementation. With thousands of highly successful ITSM implementations worldwide, ServiceNow is uniquely positioned to provide prescriptive guidance with regard to implementation, upgrades, and value realization within the context of these industry best practices. As a customer-driven organization, we continually build best practice learnings into the product while delivering supplemental guidance using a variety of channels (e.g., Customer Success Center, ServiceNow Community, eBooks, white papers, podcasts, etc.) so that customers can consume it based on individual preferences.

Frameworks such as the Information Technology Infrastructure Library (ITIL), VeriSM, The Open Group Architecture Framework (TOGAF®), the IT4IT Reference Architecture (IT4IT), Control Objectives for Information and Related Technologies (COBIT), DevOps, and Scaled Agile Framework (SAFe) initiatives are often marginalized in the absence of an integrated and tailored approach that accounts for the technology stack. Additionally, it is a common misconception to think of these frameworks/methodologies in any way other than complementary.

Combining a series of framework elements with your ITSM strategy, a Digital IT Operating model and the ServiceNow technology stack can modernize and transform your entire IT ecosystem.

Combining a series of framework elements with your ITSM strategy, a digital IT operating model and the ServiceNow technology stack can modernize and transform your entire IT ecosystem. Precision selection and alignment of the relevant framework elements along with the ServiceNow technology stack will accelerate your journey to the digital enterprise. *There will be considerations that are specific to your organization but the high-level guidance is to use the architecture review board (ARB) and architectural principles of TOGAF, the service management principles of ITIL, the service backbone and common data model from the IT4IT standard, the digital transformation perspectives of VeriSM, the agile concepts from the Scaled Agile Framework (SAFe), and the culture that thrives within DevOps.*

The term “reference architecture” has a degree of ambiguity across the industry. For the purpose of this eBook, reference architecture is defined as a collection of recommended best practices as they relate to the Now Platform and ITSM applications. The ServiceNow practitioner community has extensive knowledge when it comes to ITSM implementations and maximizing the return on investment. The ITSM Reference Architecture embraces the diversity of the various frameworks so customers can launch an ITSM modernization journey that aligns to their IT strategy and delivers the desired business outcomes.



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ITSM strategy calibration

The ideal starting point is to periodically recalibrate your ITSM strategy. A clearly documented strategy combined with the modern ServiceNow technology stack will serve as an accelerator when it comes to digital transformation and time to value. ITSM should be viewed as a strategic asset in the organization that exponentially increases business value at every stage of the journey. Many ITSM programs lose momentum after the initial implementation of the most common ITSM applications such as Incident Management, Service Desk, and Change Management.

Having a well thought out ITSM strategy that includes a digital transformation roadmap will ensure that your ITSM program continually adds incremental business value with each newly activated ServiceNow capability and release. Oftentimes, ITSM strategy is dismissed as theory or viewed as simply an academic exercise. The significance of strategic planning is best articulated by the following ITIL statement: “Theory is the very basis of best practice.” A well-documented strategy combined with a digital transformation roadmap is essential when it comes to realizing the full value of your ServiceNow investment.

It is recommended that your ITSM program have a well-documented strategy. It should describe the underlying principles and design considerations necessary to correctly position and deliver value to the business. The “service strategy” lifecycle is represented as the nucleus of the ITIL framework image shown to the left. ITSM leaders should use their documented strategy as a “check and balance” for all ITSM program decisions.

Within the strategy, it is recommended that a series of supporting strategies be developed for each ITSM service provider. For example, there should be individually documented strategies for your service desk, change management, etc. However, great care should be taken to ensure those supporting strategies do not conflict with the overarching ITSM strategy or IT strategy. If there is a potential conflict, it should be elevated to your organization’s ITSM architectural review board (ITSM-ARB) for consultation. Strategic plans should also include the organization’s purpose along with a clear statement of direction. This can be in the form of a vision statement, mission statement, or simply a tagline (e.g., Nike’s “Just Do It”). The statement of direction serves as an anchor for program decision-making, roadmap development, and value realization.

The organization's ability to transform grows exponentially when everyone understands their role in the value chain.

Activity 1: Strategic planning

There are four fundamental design perspectives when it comes to strategic planning exercises. They are (1) vision-based, (2) issue-based, (3) alignment-based, and (4) scenario-based. These strategic planning methodologies typically share a common set of characteristics that can be tailored to the organizational objectives. Frequently, a combination of approaches is used throughout the program, product, or service lifecycle to recalibrate and maintain alignment to changing market conditions, address specific problems, and/or chart a series of scenarios that could potentially play out as a competitive advantage.

Strategic planning doesn't have to be complicated, but it does involve more than just brainstorming and creating a strategy document with limited distribution. It involves (1) articulating the vision and direction, (2) expressing distinctiveness, (3) documenting the transition/journey, and (4) identifying a continuum of actions that will be necessary to achieve the stated objectives. The output should be guidance that can be consumed by everyone in the IT ecosystem. The organization's ability to transform grows exponentially when everyone understands their role in the value chain.

These principles should be applied across the ITSM ecosystem from top to bottom. Every internal and external service provider should have a clear understanding of their role in the value chain. This can be difficult at times especially when it comes to expressing distinctiveness. There are cases when an internal service provider may have to acknowledge that an external provider can deliver a set of capabilities faster, better, and/or cheaper. In the final execution plan, the ITSM delivery team should be aligned from top to bottom and be incentivized to make the best decision for the organization. The plan should include specific guidance on people, process, technology, architecture, and capability activation.

Activity 2: Crafting an ITSM statement of direction

Creating and publishing the ITSM statement of direction (e.g., vision, mission, or mantra) can have enormous value. Mission describes the fundamental purpose of the program while vision describes the intended future state. When crafting these, it can be valuable to perform a quick Strengths, Weaknesses, Opportunities, and Threats (SWOT) analysis and examine vision, mission, and mantra examples from other companies. Keep in mind that the public vision and mission statements of companies are often very loosely written by design (e.g., commander's intent, which succinctly describes what constitutes success without defining the exact steps needed to achieve it), but internal vision and mission statements for the ITSM program should be more concise. In the end, you want all the elements of the strategic plan to be aligned and understood by everyone in the organization.

When the ITSM statement of direction is clear, it should be very easy for both internal and external service providers to tailor their engagement models, decision making, and performance objectives to support the desired business outcome.

Vision	Describes the organizations desired or future state—where you want to be at some point in the future	ITSM vision statement (example): <i>Our ITSM program will serve as a strategic asset and key differentiator for our organization with a continual focus on optimizing service delivery, improving stability, driving down costs, and driving standardization throughout the enterprise by using the breadth of ServiceNow capabilities to modernize our ITSM program.</i>
Mission	Describes organization's fundamental purpose and current approach to achieving the mission	ITSM mission statement (example): <i>Our ITSM program will use ServiceNow for IT service management (ITSM), IT governance, enterprise architecture, and information security to develop a service delivery model that adds value and serves as an enabler to both IT and the business. Additionally, the entire program will be underpinned by the Deming Cycle, often referred to as the Plan-Do-Check/Study-Act (PDCA) Cycle, to ensure the program always remains poised to enable the business.</i>
Tag lines	Meant to provide consumers with an indication of your brand and its market position in just a few memorable words	Tag lines (examples): <i>Modernizing the way we work, delightful customer experience, single system of engagement.</i>

Herb Kelleher, the longest serving CEO of Southwest Airlines, once said, "I can teach you the secret to running this airline in thirty seconds. This is it: **We are THE low-fare airline.** Once you understand that fact, you can make any decision about this company's future as well as I can."

Having a vision of something like "being the market leader" gets everyone energized; having a detailed strategic plan that frames up the journey allows the organization to achieve it.

Activity 3: Defining success criteria

Outlining specific success criteria as part of strategy development is essential. The success criteria should be based the well-known SMART (Simple, Measurable, Achievable, Relevant, and Timed) goals and they can be tiered in nature to secure short-term wins and build confidence for the team with respect to the strategy. These can also be incorporated into individual performance plans so that everyone is always working toward realization of the vision. Take a moment to document a few items that might qualify as success criteria for your ITSM program today.

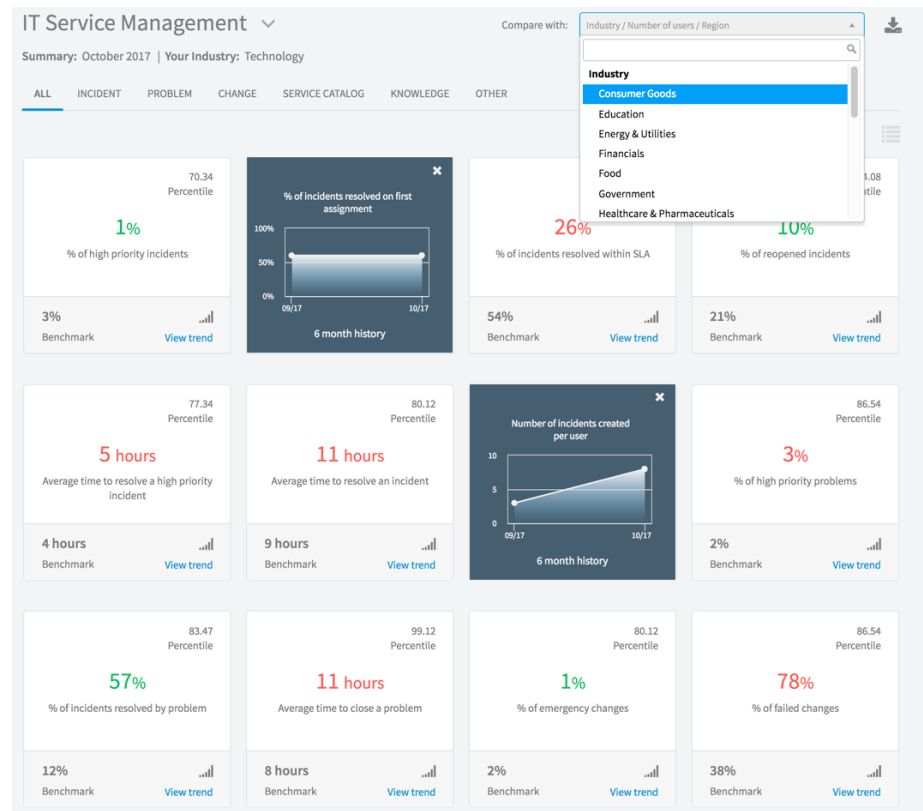
Activity 4: measuring success

Once the success criteria are defined, it will be necessary to define key performance indicators (KPIs) that underpin the success criteria/goals. Setting KPI targets (e.g., % increase, % decrease of a metric) can be very powerful when it comes to modifying behaviors and achieving success.

While there are likely to be some KPIs that are specific to your strategy, ServiceNow exposes the most common ITSM KPIs via ServiceNow Benchmarks. These benchmark KPIs are incredibly powerful when it comes to understanding your performance relative to like customers (e.g., by industry, by geography, or by size). KPIs are most commonly used to shape the behaviors around a certain metric. The out-of-the-box (OOTB) KPIs delivered via ServiceNow Benchmarks can be seen on the following page.

Setting KPI targets (e.g., % increase, % decrease of a metric) can be very powerful when it comes to modifying behaviors and achieving success.

Each ServiceNow Benchmark KPI also includes a set of ranked recommendations for KPIs that are performing below the global benchmark. This is an incredibly powerful capability that can be used to drive behavior change and visualize the results of that modified behavior or corrective action.



It is important that you measure success and monitor and track your journey. ServiceNow includes a wide range of reporting and analytics capabilities so that our customers can make informed business decisions about their ITSM program.

ITSM governance strategy

Governance, risk, and compliance are a constant focus area for all enterprises. It is imperative that IT keep up with changes in the global regulatory environment and industry standards to protect the business. As enterprises adopt new business models, establish new partner relationships, and deploy new technologies, they must also quickly assess the impact of these developments on their existing compliance obligations and risk posture.

Similarly, it is advised that you establish architecture and governance oversight of the ITSM program. It is recommended that ServiceNow customers use a variant of the TOGAF-defined architecture review board (ARB) to oversee both architecture and governance decisions related to the ITSM program. Combining this ARB model with some of the key governance concepts offered in the Control Objectives for IT (COBIT) framework and ServiceNow GRC can have tremendous value. Specifically, following the Governance of Enterprise IT (GEIT) guidance to align IT goals with specific business objectives is highly recommended. There are some examples in the mapping exercise in the section on creating the ITSM roadmap (p. 25).

Establishing an ITSM-ARB is important for both early journey and mature customers.

Activity 1: ITSM architecture review board (ITSM-ARB)

An ITSM architecture review board (ITSM-ARB) is recommended for governance, risk, and compliance oversight of the ITSM program. The creation of an ITSM architecture review board, ITSM-ARB, is key to any IT governance strategy. It is a cross-functional team that oversees and governs the ITSM program. The ITSM-ARB should have a charter that incorporates ITSM strategy, governance, roadmap, policy, customization management, industry best practice alignment, security, and architecture. Additionally, this should align with the overarching IT governance, risk, and compliance strategy of the organization.

Establishing an ITSM-ARB is important for both early journey and mature customers. Customers that are actively transitioning “back to baseline” will benefit by leveraging the ARB to protect all their “back to baseline” efforts against future drift. While early journey customers benefit by proactively managing drift from the point of their “go live.” The initial steps of setting up an ITSM-ARB include appointing an executive sponsor, assigning an ITSM-ARB chairperson, creating a charter, and identifying membership. This doesn’t have to be a complicated set of activities or a costly investment.

Activity 2: Customization versus configuration

Traditionally, customizations have been viewed as carrying both excessive cost and risk while configuration changes are viewed as low risk. A more modern viewpoint is to think about both configuration and customization in the context of technical debt. The Now Platform is the ideal development platform for custom applications, and there may be a perfectly valid business reason for making an OOTB application customization. Since ITSM continually incorporates field-tested best practices into the product natively, there should not be a lot of cases where it is advisable to customize those OOTB ITSM applications, but the point is that not all customizations are risky and conversely not all configurations are safe.

The recommended perspective is to think about all modifications (customization or configuration) in terms of technical debt. The attributes of technical debt include upgradeability, manageability, and supportability. By taking a look at any modification through this lens, informed decisions can be made about the best implementation solution based on the technical debt that solution will incur (cost). That cost can then be weighed against the business outcome to determine if the modification is worth the overall cost. One other recommendation is to think about the modification in terms of all customers and periodically review the ServiceNow ITSM release roadmap with your account team. ServiceNow is a customer-driven organization so share your ideas and needs with us so we can help.

ServiceNow continues to incorporate field-tested best practices into the (OOTB) applications while leading the industry with new innovation. Legacy ITSM transition customers should make every effort to maintain close alignment with ServiceNow OOTB ITSM applications. This drives fast time to value, lowers TCO and simplifies upgrades. As ServiceNow releases new capabilities, ITSM service providers can quickly deliver those new capabilities to their business users.

ITSM modernization initiatives typically have the greatest success when they are built to serve as an integral part of the business value chain.

Best practices for evaluating technical debt

- Consider the risk versus reward of the proposed change in terms of upgrades, supportability, and maintainability.
- Use the Now Platform functionality whenever possible.
- Treat customizations just like you would any SW engineering work.
- Exercise good practices when it comes to SW engineering principles.
- Use the ServiceNow Automated Test Framework (ATF) where possible.
- Make sure that functionality works in both sub-prod and prod after an upgrade
- Make a duplicate of the code that is to be customized.
- Make the original record inactive—this should be the only update that is made to the original record.

While there may be hundreds of customizations in legacy tool sets, it is important to carefully evaluate all customizations from a technical debt perspective before porting them to the new modern ITSM applications in ServiceNow. The recommendation is to evaluate process requirements upfront and make sure that a legacy customization is something that is absolutely required to add business value. Legacy processes and procedures are often outdated or no longer needed so the related customizations should be evaluated using something like the “5 Whys” to determine if it is still necessary. All customizations should be documented along with the underlying business drivers in ServiceNow Knowledge Management or the SW documentation library of choice.

Activity 3: Modernizing ITSM

As a ServiceNow customer, you have already taken the most important step when it comes to modernizing the way people work and accelerating your IT digital transformation journey. Digital transformation requires a paradigm shift in thinking when it comes to automation, customer experience, and serving the business. ITSM modernization initiatives typically have the greatest success when they are built to serve as an integral part of the business value chain.

ServiceNow is uniquely positioned to help customers realize the full return on their ITSM investment. But it is important to note that modernization with ServiceNow extends well beyond ITSM. Modernization with ServiceNow creates a paradigm shift in the way people work and how value is delivered to the business.

For the ITSM program to serve a strategic asset and become a competitive advantage for the business, it is important to fully understand the desired business outcomes and preferred consumption models so that the ServiceNow instance is tailored to the specific needs of your business. Tailoring services to the needs of the business often starts with mapping the desired outcomes to the underlying capabilities included in the Now Platform. Capabilities are simply a means for achieving outcomes that either enable or create value for the business. A quick capability mapping exercise using the COBIT Governance of Enterprise IT guidance to align IT with specific business objectives is highly recommended. This exercise allows the ITSM team to visualize their role and contribution to the business value chain.



Important note: There is no single journey that can be applied to all customers, but the following steps can help frame up a value realization journey that serves your organization.

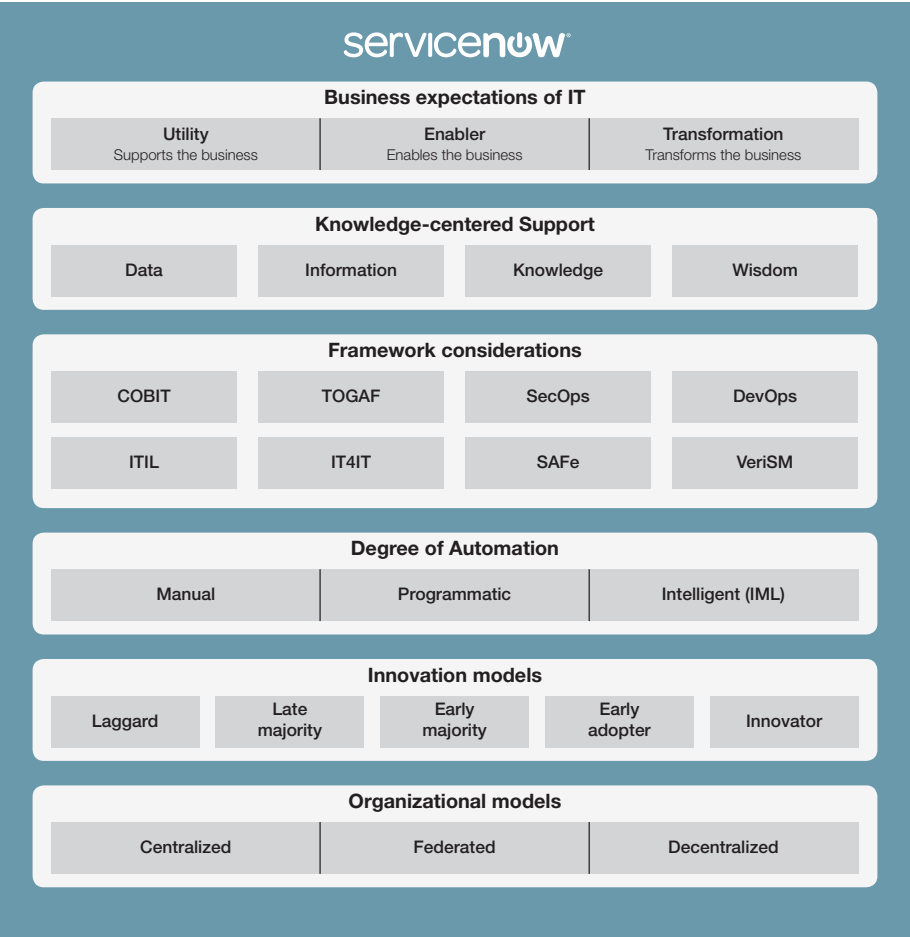
Aligning the digital transformation operating model with your strategy is essential for successful ITSM planning, roadmap development, and building out the customer journey.

Digital transformation operating model

Most IT leaders have an excellent understanding of their organization’s IT service delivery ecosystem and already have initiatives underway to digitize their operational capabilities. However, periodic recalibration of the operating model is necessary due to the rapid pace of technology and the continual evolution of the digital enterprise. IT has to remain agile and competitive in today’s fast-paced business environments. In most cases, the ideal operating model blends elements of DevOps, ITIL, IT4IT, TOGAF, SAFe, VeriSM, and COBIT into an IT service delivery ecosystem that is tailored to the needs of the business. Each of these best practice frameworks and methodologies have strengths and weaknesses, so it is important to very intentionally architect the digital operating model to the needs of your organization. At the start of the eBook, we outlined a series of recommended elements but it will be important to tailor those to your business.

Aligning the digital transformation operating model with your strategy is essential for successful ITSM planning, roadmap development, and building out the customer journey. In a previous section, a recommended set of framework elements were identified, but all customer journeys are slightly different. When evaluating the operating model, note the loose vertical correlation between the various layers of the digital operating model.

Customer journeys often become fragmented and lose momentum over time so periodically recalibrating and re-energizing is vitally important. The various tracks contained in the model below can be used to help identify the current and target positions of your ITSM program. Ultimately, you want to review all the tracks below and then publish a tailored digital operating model that is specific to your organization.



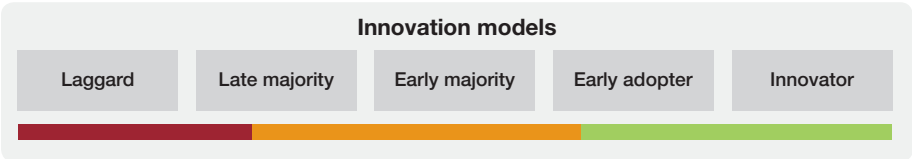
Maintaining an “early adopter” or “innovator” position in your ITSM operating model can have enormous benefits when it comes to value realization with ServiceNow.

In the next few sections, you will navigate through a series of planning activities that are designed to flesh out a digital IT operating model that is tailored to your organization. This can be used for many purposes and should be fully aligned to your ITSM strategy.

Activity 1: The innovation model

When it comes modernization, one of the most important considerations is placement on the innovation scale (see below). This includes your ServiceNow release adoption strategy. Maintaining an “early adopter” or “innovator” position in your ITSM operating model can have enormous benefits when it comes to value realization with ServiceNow.

ServiceNow typically releases new features and capabilities twice a year so maintaining an early adopter or innovator position accelerates time to value. As new capabilities are released by ServiceNow, customers can immediately use those new capabilities and pass them on to their business users. Modernization becomes a native part of your ITSM program by simply getting in position to consume new releases from ServiceNow as quickly as possible.



One of the critical success factors for maintaining the early adopter positions is to keep technical debt to a minimum. This generally translates to avoiding customizations or configurations that translate to significant technical debt. Any custom application development should be done as a ServiceNow scoped application to minimize technical debt. We will cover scoped applications later in the document. If you have a ServiceNow instance that already has a lot of technical debt, incorporating a debt reduction strategy or back to baseline initiative is recommended.



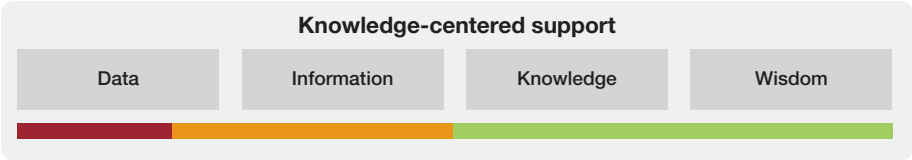
Important note: The innovation model should include a classification mechanism that allows you to assign certain risk values or categories by application. This may be based on their quality history and other attributes. Providers that consistently deliver high quality applications would be slotted differently than applications from providers that have a less than stellar quality record. Therefore, you might place ServiceNow in the “innovator” slot and application XYZ in the “late majority” slot.

Activity 2: knowledge-centered support (KCS)

Knowledge is one of the most powerful yet undervalued elements of the digital operating model. Whether it is incident deflection, how-to instructions, or rapid resolution, knowledge can be incredibly powerful. Knowledge improves customer satisfaction, lowers operating costs and reduces human error.

Combining ML and AI with KCS can totally change the landscape of ITSM.

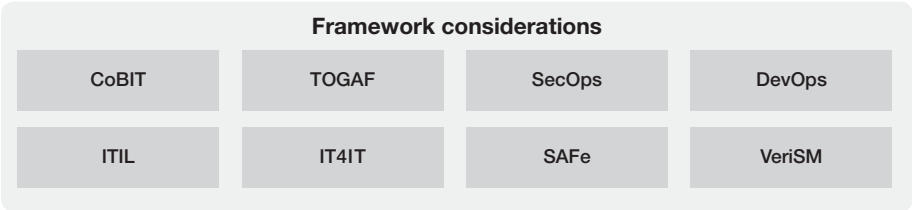
Recently, innovative organizations like ServiceNow have started to stretch the boundaries of targeted knowledge by introducing machine learning (ML) and artificial intelligence (AI) into the platform capabilities. While machine learning and artificial intelligence will never displace the power of the human brain or the IT workforce, there are areas within IT where these capabilities can add enormous value. They can improve speed and accuracy and lower costs. Combining ML and AI with KCS can totally change the landscape of ITSM. The chart below shows the organic transition from data to information to knowledge to wisdom. The ideal position from a digital operating model perspective is a position somewhere between knowledge and wisdom.



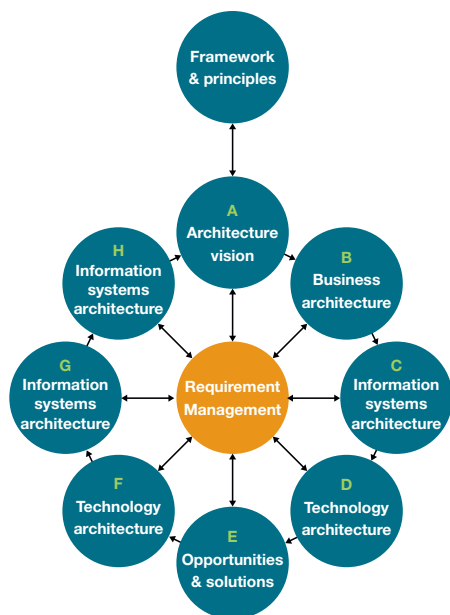
Crowdsourcing, communities, and knowledge valuation are also key considerations when it comes to KCS positioning in the digital operating model. Knowledge management is much more than the tactical management of knowledge articles. In this context, KCS should be viewed from an investment strategy perspective. Research shows that there are clear areas where knowledge articles are heavily used and reused. Consider the possibility of using machine learning to identify the best places to make your knowledge investments and then how that knowledge could potentially be presented back to your customers. This is an area that is far from reaching its full potential.

Activity 3: Industry framework and methodology alignment

There are numerous industry frameworks, methodologies, practices, and standards that are applicable to the practice of ITSM. Many of these are abstracted away from the technology by design so that they can be applied to any organization or technology stack. This is an ideal model from a framework or methodology perspective, but it leaves customers with the challenge of interpreting the framework for implementation purposes. Customers are often forced to make large investments in human capital to practically interpret and implement these frameworks. Even that is often subject to interpretation and significant variation depending on the technology stack and/or practical implementation expertise of the implementer. It is rarely the case when a singular framework approach will serve the organization well. Below is a quick overview of the most common frameworks and standards related to the practice of ITSM along with a ServiceNow point of view (POV).



Important note: ServiceNow strives to be framework neutral by selectively embracing high-value elements in each of these that have a proven track record of success. The recommendation is to incorporate elements of each of these frameworks into a single operating model that serves your organization.



The Open Group Architecture Framework
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TOGAF

The Open Group Architecture Framework, TOGAF, is an Open Group standard that is widely recognized as the de facto standard for enterprise architecture. It is both a methodology and a framework that is used by the world's leading organizations to improve business efficiency. The Open Group Architecture Forum continues to update the framework and methodology on regular intervals to ensure it remains relevant.

ServiceNow TOGAF point of view (POV):

Maintaining close alignment between service management and enterprise architecture has enormous value. There are concepts across the different disciplines that complement each other and strengthen the value propositions of both enterprise architecture and service management.

It is recommended that you use a common architecture repository that contains Architecture Building Blocks (ABB), Solution Building Blocks (SBB) and Process Building Blocks (PBB), to standardize architectures, accelerate initiatives, and drive down costs. Nesting building blocks and loosely coupling them whilst exposing important services ultimately reduces the overall complexity and improves time to market with new process implementations.

It is also recommended that customers use an ITSM-focused variant of the architecture review board (ARB) that is described in TOGAF. The recommended tailoring activity is to narrow the focus and stand up an ITSM architecture review board (ITSM-ARB). It should serve as cross-organization body that oversees and governs the ITSM program. The ITSM-ARB should have a charter that incorporates ITSM strategy, governance, roadmap, policy, customization management, industry best practice alignment, security, and architecture. Additionally, this should align also with the overarching IT governance, risk, and compliance strategy of the organization.

For more information on TOGAF, visit:

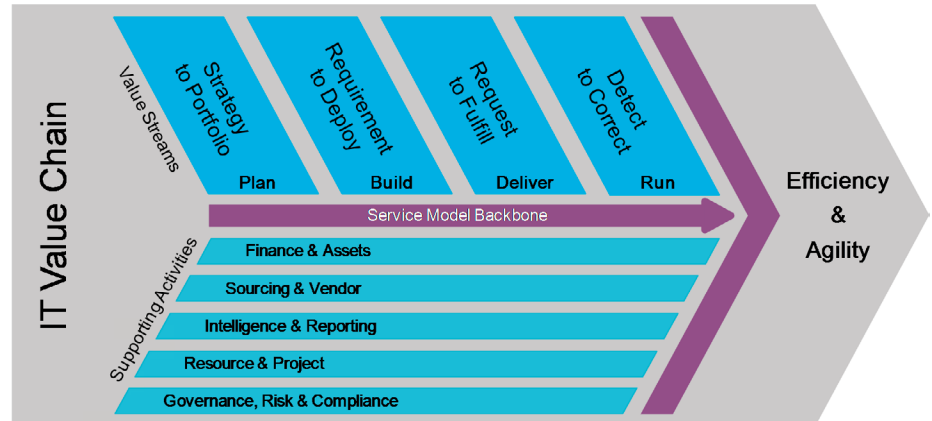
opengroup.org/subjectareas/enterprise/togaf

IT4IT

The Open Group IT4IT organization provides a standard reference architecture for managing the business of IT. It uses a value chain approach to create a functional data model for IT value creation. The IT4IT standard provides a vendor-neutral, technology-agnostic, and industry-agnostic reference architecture for managing the business of IT, enabling insight for continuous improvement.

The IT4IT Reference Architecture standard consists of the IT value chain and a three-layer reference architecture. The IT value chain is made up of the four IT value streams, which can play a vital role in helping IT control the service model as it advances through its lifecycle. The four IT values streams are Service to Portfolio (S2P), Requirement to Deploy (R2D), Request to Fulfill (R2F) and Detect to Correct (D2C).

IT4IT should be considered complementary to the other frameworks and methodologies such as ITIL and VeriSM.



IT4IT value chain

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ServiceNow IT4IT point of view (POV): IT4IT is most valuable for identifying capability gaps in the IT delivery model, establishing a common service backbone/data model and for running IT as a business. IT4IT can be a powerful tool when it comes to closing gaps in the service delivery model. ServiceNow fully embraces the IT4IT standard and has some of the broadest IT4IT capability coverage on the market.

IT4IT should be considered complementary to the other frameworks and methodologies such as ITIL and VeriSM. IT4IT is primarily a data-driven reference architecture model that leverages that concept of a “service backbone”. It can serve as a powerful reference when it comes to mapping out customer journeys and capability activation order but don’t view this as an alternative to ITIL.

For more information on IT4IT, visit: opengroup.org/it4it/about

Information Technology Infrastructure Library (ITIL)

ITIL is a set of detailed practices for ITSM that focuses on aligning IT services with the needs of the business. In its current form (v3 2011), ITIL is published as a series of five core volumes, each of which covers a different ITSM lifecycle. Although ITIL uses ISO/IEC 20000, the International Service Management Standard for IT service management, as reference model for certification purposes, there are differences between the ISO/IEC 20000 standard and the ITIL framework.

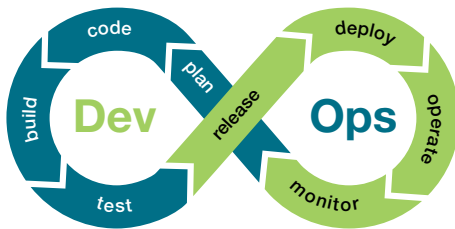
DevOps is more of a culture than an actual framework.

ServiceNow ITIL point of view (POV): ITIL is the de facto standard for most ITSM programs and it has a number of sound principles that have been widely adopted by IT. It has five lifecycles (Service Strategy, Service Design, Service Transition, Service Operation, and Continual Service Improvement). Those lifecycles represent 26 processes and four functions.

The principles of ITIL have been largely adopted in the market along with most of the Service Operation lifecycle. The other lifecycles have pockets of widespread process adoption (e.g., Change Management, Knowledge, SACM, etc.) but haven't had the same traction from an overarching lifecycle perspective.

However, the ITIL model is a set of best practice guidance and therefore not intended to be taken literally. In fact, the way to achieve process maturity in the world of ITIL is to tailor those best practice models to your organization. While there has not been a content refresh since 2011, AXELOS is leading a team of practitioners on a refresh exercise.

For more information on ITIL, visit: axelos.com/best-practice-solutions/itil



DevOps

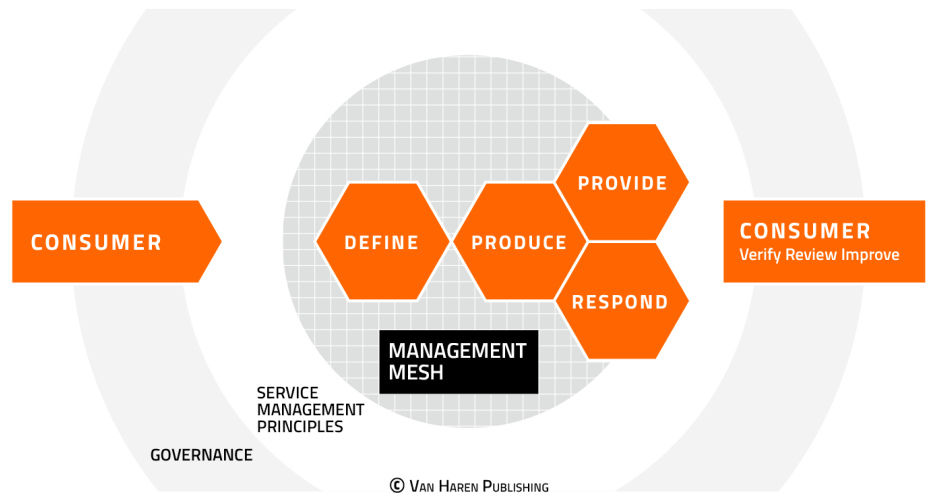
DevOps is more of a culture than an actual framework. It incorporates a variety of philosophies, techniques, practices, and delivery models that are intended to unify development (dev) and IT operations (ops). There are series of frameworks and methods that can serve as an overlay for DevOps such as the SAFe. However, the fundamental idea is that velocity and agility help organizations better serve their customers and compete more effectively in the market.

Enterprise DevOps: Enterprise DevOps requires a modern set of workflows to incorporate a variety of philosophies, techniques, and practices. These modern workflows are intended to unify development (dev) and IT operations (ops). Enterprise DevOps embraces a wide range agile methods that help increase velocity. The Enterprise DevOps approach can vary from organization to organization but it typically includes support for backlog grooming, development, unit test, and continual integration. The fundamental idea is that velocity and agility help organizations better serve their customers and compete more effectively in the market.

VeriSM

VeriSM encourages service providers to see their capabilities as part of an overall organization, not as a web of internal providers and internal consumers. People, departments, and teams will work together as part of this, but as colleagues, not through a consumer-service provider relationship. Staff within different business capability areas should see other staff in the organization as their **partners**—everyone working together to deliver outcomes and value to consumers. A culture of communication and cooperation must exist to successfully support product or service provisioning, otherwise the success of service management is compromised.

VeriSM is an emerging methodology that incorporates a very modern approach to service management.



VeriSM

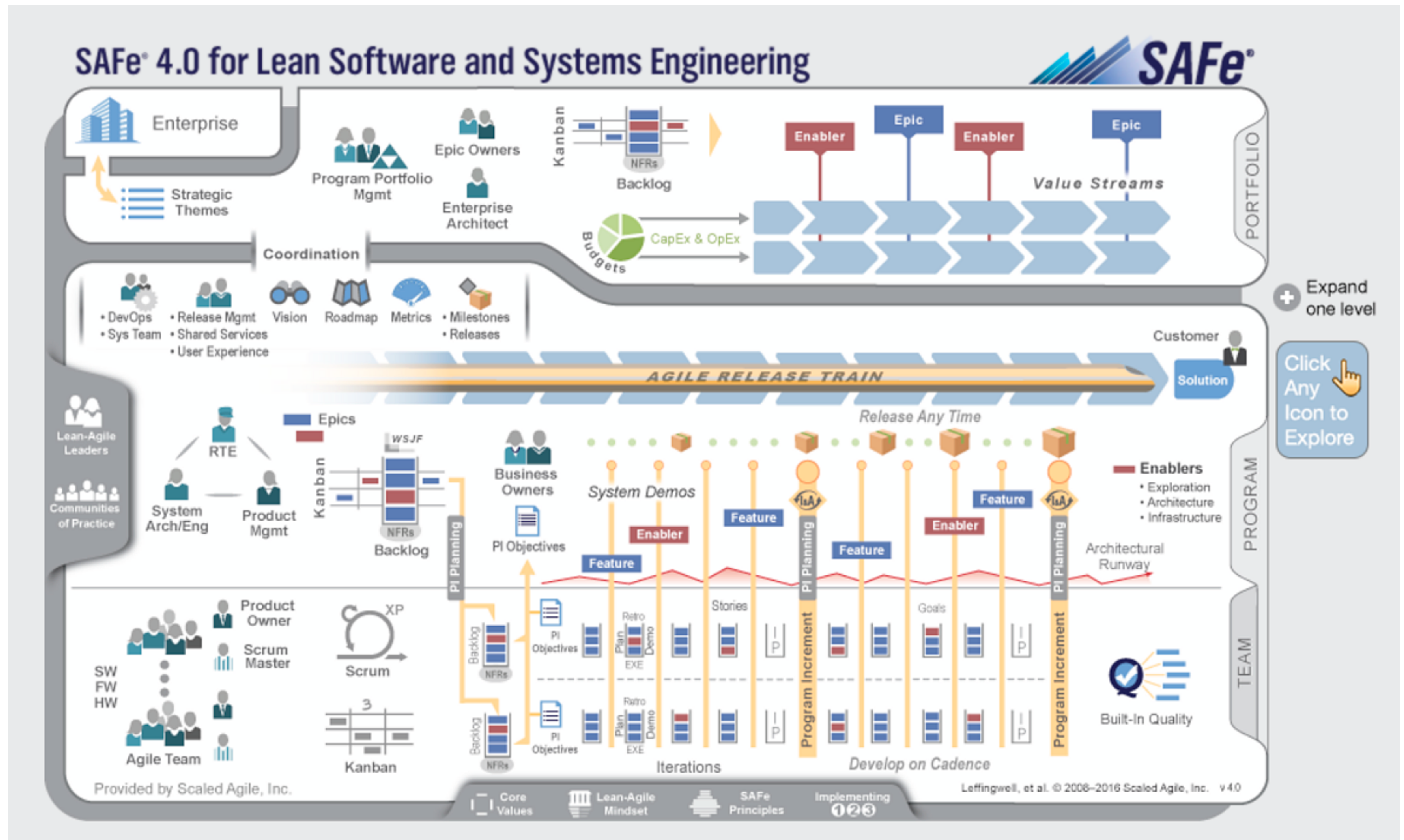
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ServiceNow VeriSM point of view (POV): VeriSM is an emerging methodology that incorporates a very modern approach to service management. The recommendation is to keep a close eye on this as it develops and adopt some of the digital perspectives. Many of the areas described in this eBook have perspectives similar to VeriSM so we are optimistic about the future of VeriSM.

For more information on VeriSM, visit: verism.global/

Scaled Agile Framework Enterprise (SAFe)

SAFe is a framework for Lean-Agile development practices. It provides guidance using a series of top-down tracks that span the Portfolio, Solutions, Program, and Teams levels. At the Program level, the team identifies the Epics of the portfolio to construct a value stream. The next phase is the creation of solutions or trains, which are constructed using raw capabilities. The teams then move into program increment (PI) planning to create Agile Release Trains (ARTs). This is the negotiation phase where the teams commit to deliver certain stories as part of the release. It is also important to note that there is a SAFe essentials path that is designed for early adopters to simplify adoption.



Scaled Agile Framework (SAFe)

Copyright © Scaled Agile Inc.

ServiceNow SAFe point of view (POV): SAFe has earned the position of being the de facto standard AGILE framework. SAFe is a very flexible and scalable framework but our recommendation is to begin with Essential SAFe, which is the most basic configuration of SAFe, and then build from there. This allows the organization to begin adopting and standardizing on key agile concepts. As the organization matures, SAFe can be upleveled to Portfolio SAFe, Large Solution SAFe and Full SAFe as needed. The concepts of SAFe are sound and we recommend adopting the models that make sense for your organization and maturity level.

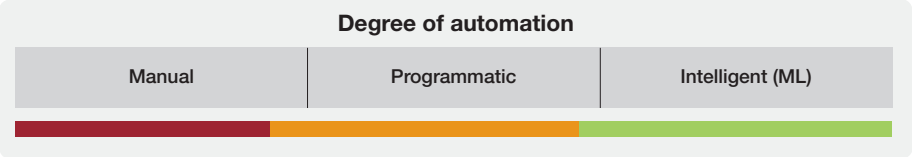
For more information on SAFe, visit: scaledagileframework.com/about/

The overarching ServiceNow recommendation is to review these and adopt the elements of each that support your strategy and program positioning. The best result is to architect a framework that is tailored to your organization and strategy. It will likely contain elements of each of the industry frameworks that manifest themselves in the form of capabilities. From the ServiceNow perspective, we embrace them all.

A ServiceNow administrator that has the skills to create a report in ServiceNow already has the skills to implement Agent Intelligence.

Activity 4: Intelligent automation

The degree of automation is a key consideration when building out a digital operating model. There is a direct correlation between the degree of automation and the digital enterprise. The more an organization shifts to the right of the automation scale below, the greater their ability to respond quickly and decisively to customer demand and errors in the infrastructure at a much lower cost. The cost reductions associated with using machine learning and artificial intelligence are realized via operational efficiencies, rapid resolution of outages, and reduced handling costs. When people can move into more fulfilling work and assignments that add more value to the organization, everyone wins.



ServiceNow's orchestration and machine learning capabilities are designed to be simple and easy to use. Our engineers do the heavy lifting so that our customers can be up and running in hours. A ServiceNow administrator that has the skills to create a report in ServiceNow already has the skills to implement ML. Here are a couple of best practices and recommendations that can help customers prepare to maximize their investment in ML/AI.

ServiceNow ML candidate qualification

Step 1: Identify problem to solve

Step 2: Identify success criteria

Step 3: Identify ML candidates (e.g., categorizations, assignment groups, knowledge)

Step 4: Examine integrity of candidate data sets

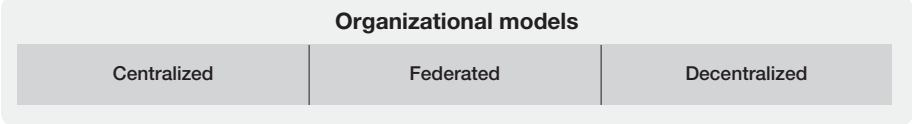
Step 5: Create plan to normalize and cleanse data sets

Step 6: Update data input rules for continued normalization of inbound data

Step 7: Train and test prediction accuracy

Activity 5: Organizational modeling

Organizational models are likely outside the control of IT so this is really an exercise in factoring organizational structure into the digital operating model. It is very common to see a hybrid of centralized, decentralized, and federated in today's enterprise environments. For that reason, we did not include a specific recommendation or target band. However, it is important to note that for Service Integration and Management (SIAM) architectures, this is a key consideration when building out the digital operating model. It often requires a very specific management strategy and reporting overlay.

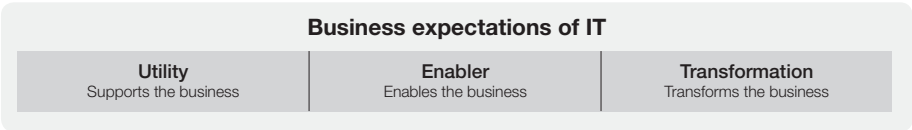


The degree of automation required may be significantly higher in a utility-based model.

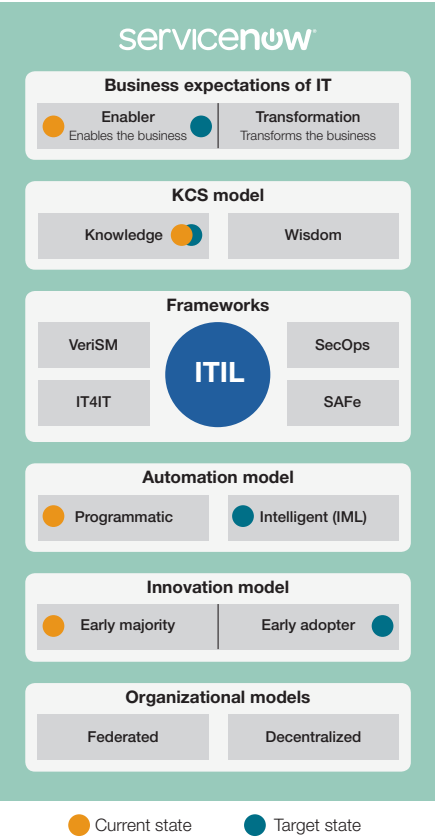
Activity 6: Business expectations of IT

There are typically one of three primary business expectations of IT. **(1) Utility** – IT is fundamentally thought of as a cost center and charted with simply supporting the business, **(2) Enabler** – IT enables the business with technology to achieve the business objectives and **(3) Transformation** – IT leads the business with innovation and uses technology for competitive advantage.

On the surface, this may seem informational, but it plays a role in the operating model definition and the financial allocation model. The degree of automation required may be significantly higher in a utility-based model. And while the base cost footprint may seem smaller in that Utility category, it could actually prove to be costlier per interaction without automation.



Important note: A fragmented operating or service delivery model often results in cost inefficiencies, complexity, and a suboptimal consumer experience. There are no two customers that are exactly alike, and an operating model reference helps optimize ITSM in the context of the business expectations. Ideally, ITSM should serve as a strategic asset that can be used for competitive advantage.



Activity 7: Tailoring the digital operating model

Now that you have assessed the current state and future state of each individual scale, it is time to pull them together and frame up a tailored digital operating model for your organization. This can serve as a powerful decision-making and communication tool as you transition into the organizational change elements of the journey. It is important to have the entire service provider ecosystem understand the digital operating model that the organization is driving to achieve. Additionally, communication around the digital operating model and the value it provides should be incorporated into an ongoing awareness campaign (see diagram at left).

ITSM maturity

Activity 1: Setting maturity targets

Identifying ITSM maturity targets and defining an assessment strategy is recommended to ensure that the organization properly allocates and manages ITSM investments. Not every ITSM program or process should strive to have a “level 5 – fully optimized” maturity rating. For example, consider two companies in different verticals. The first is a large financial institution and the second is a technology startup. They both use the same ITIL-based change management process but there is likely more rigor and discipline required for the financial institution, while the startup is likely to

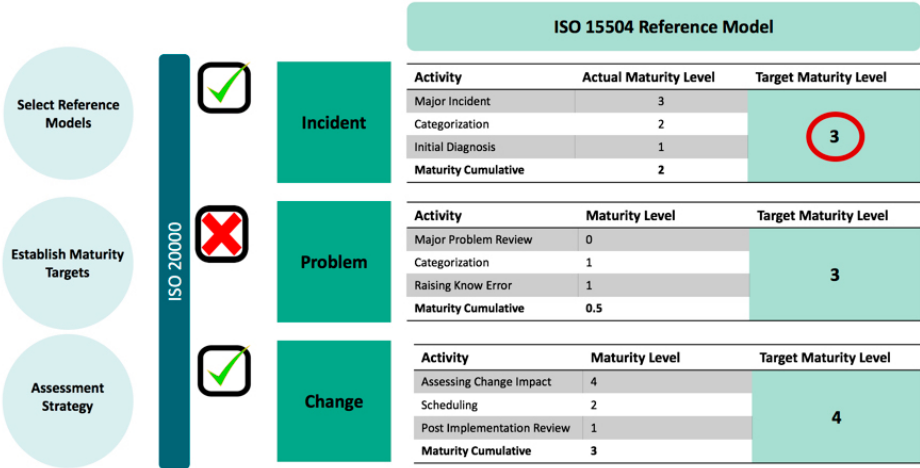
This hybrid approach can be very powerful in terms of assessing, maturing, and identifying ecosystem gaps.

have a greater appetite for risk.

Nearly all companies strive for high velocity but they must balance that with regulatory and compliance considerations for their organization. Fortunately, ServiceNow is uniquely positioned to offer customers the flexibility to create tailored workflows that allow customers to balance rigor and velocity based on their individual needs and compliance considerations.

It is often helpful to establish a baseline of individual process maturity using ISO15504 and then use ISO20000 (ISO-20K) or IT4IT for a comprehensive ITSM coverage view. This hybrid approach to maturity has numerous advantages over using a single maturity reference model. First, ISO15504 includes the ability to deep dive into individual process maturity and identify specific areas for improvement on an activity-by-activity basis, while the ISO20K perspective presents a binary view of the entire ITSM program. While IT4IT is not a maturity reference model, it does present some similarities with ISO-20K when to comes to assessing capability coverage.

This hybrid approach can be very powerful in terms of assessing, maturing, and identifying ecosystem gaps. This approach also delivers consistency and repeatability with respect to future maturity assessments, regardless of whether they are performed



ITSM Maturity

as internal self-assessments or external third-party assessments.

Activity 2: Selecting maturity reference models

ISO20000

ISO/IEC 20000, developed in 2005, is the international standard for IT service management. ISO/IEC 20000 was originally developed to reflect best practice guidance contained within the ITIL framework, although it equally supports other IT service management frameworks. ISO2000 provides a comprehensive layout view of ITIL, so it can be useful when it comes to identifying gaps in the ITSM ecosystem.

ISO/IEC 15504®

ISO/IEC 15504® is the reference model that assessors can use to consider evidence collected during their assessment to determine ratings. The assessors can give an overall determination of the organization's capabilities for delivering products (software, systems, and IT services). The ISO 15504 maturity rating scale is listed below. One of the other advantages of using ISO 15504 is that it has enough depth to remove some of the subjectivity that is typically seen in some of the high-level analyst supplied maturity models.

Rating	Level name	Level definition
0	Incomplete	There is a general failure to attain the purpose of the process. There are no easily identifiable work products or outputs of the process.
1	Performed	The purpose of the process is generally achieved. The achievement may not be rigorously planned and tracked. Individuals within the organization recognize that an action should be performed, and there is general agreement that this action is performed as and when required. There are identifiable work products for the process and these testify to the achievement of the purpose.
2	Managed	The process delivers work products of acceptable quality within defined time scales. Performance according to specified procedures is planned and tracked. Work products conform to specified standards and requirements.
3	Established	The process is performed and managed using a defined process based upon good principles. Individual implementations of the process use approved, tailored versions of standard and documented processes. The resources necessary to establish the process definition are also in place.
4	Predictable	The defined process is performed consistently in practice, within defined control limits, to achieve its goals. Detailed measures of performance are collected and analyzed. This practice leads to a quantitative understanding of process capability and an improved ability to predict performance. The quality of work products is quantitatively known.
5	Optimizing	Performance of the process is optimized to meet current and future business needs, and the process achieves repeatability in meeting its defined business goals. Quantitative process effectiveness and efficiency goals (targets) for performance are established, based on the business goals of the organization. Obtaining quantitative feedback enables continuous process monitoring against these goals, and improvement is achieved by analysis of the results. Optimizing a process involves piloting innovative ideas and technologies and changing non-effective processes to meet defined goals and objectives.

ISO 15504® maturity rating scale

Having a recurring cadence of both internal and external assessments helps to create a healthy check and balance.

Activity 3: Maturity assessment strategy

It is recommended that you incorporate an assessment strategy into your program. Having a recurring cadence of both internal and external assessments helps to create a healthy check and balance. The recommended best practice for maturity assessments is to alternate between internal assessments and external 3rd party maturity assessments. This approach also delivers consistency and repeatability with respect to future maturity assessments, regardless of whether they are performed as internal self-assessments or external third-party assessments.

ITSM roadmap creation

The recommended next step is to incorporate maturity actuals and targets into an ITSM roadmap. This helps the organization gauge where they are in terms of maturity, where they want to be, and the time table for getting there on a process-by-process basis. In some cases, low maturity ratings may be a symptom of not having the associated ServiceNow application capability activated.

Again, every process team should rightsize the maturity level targets based on the needs of the organization. There are many cases where a “Level 2 – Managed” or “Level 3 – Established” process is exactly the right level and no further investment should be made in maturing that process. That is not to say that the organization should not continue to automate and optimize with new capabilities, but it does provide a clear direction in terms of your maturity investments.

Activity 1: Capability modeling

This activity is based on the COBIT Governance of Enterprise IT guidance for aligning IT goals with specific business objectives. It is best to begin by completing an inventory of your current ITSM capabilities. Once the inventory is complete, a ServiceNow partner, solution consultant, or PS representative can help map this to a modern ServiceNow capability. The level at which capabilities are defined varies from organization to organization, which is why this roadmap exercise starts at the raw capability level.

ITSM roadmap elements can easily be mapped to specific business goals. This is useful in setting implementation priority and ultimately defining the customer journey. This can be done by filtering capabilities through a series of industry framework filters, so that the ITSM-ARB can quickly identify the relationship with specific business goals and objectives.

It is recommended that you move the raw capabilities into well-known categories, so that they can be easily packaged. Once this mapping exercise is complete, it is simply a matter of evaluating the capabilities that underpin the various business objectives to determine exactly which ServiceNow applications should be activated next in the journey. In the example below, the business goals and objectives were categorized into four different balanced scorecard perspectives (operational, financial, strategic, and industry). These can and should be modified to the balanced scorecard perspectives

that make sense in the organization.

Let's run through a quick modeling activity to demonstrate how this might work. Start with some high-level categories such as change management, service catalog management, etc. In this example, we used ITIL as the initial categorization schema. However, you could just as easily use IT4IT. The key is to use something that will allow you to map the capabilities all the way through to the IT and business goals. The next step is to populate IT capabilities in the appropriate categories. In the image below, there is a (1) unique identifier assigned to each (2) capability name. This is useful to ensure you have traceability all the way through the mapping exercise.

Identifier	Capability Name	Process	LC	Identifier	Capability Name	Process	LC	Identifier	Capability Name	Process	LC
1	DM001 Service Forecasting	Demand Management	Service Strategy	CM001 Change Documentation and Visibility	Change Management	Service Strategy	CM002 Change Assessment	EM001 Monitoring	Event Management	Service Operations	EM002 Event Management
	DM002 Demand Patterns			CM003 Change Authorization			CM004 Change Planning	EM003 Automated Actions			EM004 Service Impact
	DM003 Service Packages			CM005 Change Post Implementation Review			CM006 Change Monitoring	EM005 Event Significance			EM006 Performance Data Collection
	DM004 Service Pricing			CM007 Change Scheduling			CFM001 Configuration Management Planning	EM007 Service Level Management			IM001 Incident Tracking
	FM001 Financial Management Foundation	Financial Management		CFM002 Configuration Management Identification			CFM003 Configuration Management Control	IM002 Incident Lifecycle Management			IM003 Trend Analysis
	FM002 Service Based Costing			CFM004 Configuration Management Status Accounting and Reporting			CFM005 Configuration Management Verification & Audit	IM004A Integrated Service Management Functions			IM004B Process Interface to EVTM
	FM003 Cost Communication			ITAM01 Service Asset Management Policy			ITAM02 ITAM Software Portfolio Management	IM004C Process Interface to Service Desk			IM004D Process Interface to RQFM
	FM004 Customer Account Tracking			ITAM03 IT Asset Portfolio Mgmt - Asset Procurement			ITAM04 Inventory Management - Asset Receiving	IM004E Process Interface to PRBM			IM004F Process Interface to CHGM
	FM005 Chargeback			ITAM05 Inventory Management - Asset Tracking			ITAM06 Inventory Management - Software License Management	IM005 Staff Optimization			IM006 Process Optimization
	AVL01 Component Availability Management	Availability Management		ITAM07 Inventory Management - Hardware Management			ITAM08 Inventory Management - Audit and Verification	IM007 Cost Model for INCM			PM001 Problem Tracking
	AVL02 Service Availability Management										
	AVL03 Availability Plan										
	CAPO1 Component Capacity Management	Capacity Management									
	CAPO2 Service Capacity Management										
	CAPO3 Business Capacity Management										
	CAPO4 Integrated Service Management Functions										
	CAPO5 Foundational Process										
	SCM001 Service Definition	Service Catalog Management									
	SCM002 Maintain a Service Catalog										
	SCM003 Service Offerings (Customer View)										

The next step is to map the (2) capabilities from the previous exercise to the (1) COBIT control objectives. The reason we recommend the combination of COBIT and ITIL is because these maps are publically available. The purpose of this exercise is to simplify the mapping to IT and business goals. This exercise also starts to surface areas that may have gaps or deficiencies.

Identifier	Control Objectives	ITSM Capability	Effectiveness	Efficiency	Confidentiality	Integrity	Availability	Compliance	Reliability
Plan and Organize									
P01	Define a strategic IT plan	ITSM305 ITSM306 ITSM401 ITSM502							
P02	Define the information architecture								
P03	Determine technological direction	ITSM401							
P04	Define the IT process, organization & relationships	ITSM101 ITSM102 ITSM103 ITSM104 ITSM105 ITSM202							
P05	Manage the IT investment	ITSM209 ITSM402 ITSM502							
P06	Communicate management aims & direction	ITSM203 ITSM205 ITSM303 ITSM306 ITSM404 ITSM502							
P07	Manage IT human resources	ITSM209 ITSM306 ITSM403 ITSM405							
P08	Manage quality	ITSM208 ITSM402 ITSM403 ITSM405							
P09	Assess & manage IT risks	ITSM101 ITSM102 ITSM103 ITSM104 ITSM105							
P10	Manage projects	ITSM206 ITSM207 ITSM405							
Acquire and Implement									
A11	Identify automated solutions	ITSM403 ITSM405							
A12	Acquire & maintain application software	ITSM209 ITSM210 ITSM403 ITSM502							
A13	Acquire & maintain technology infrastructure	ITSM209 ITSM210 ITSM403 ITSM502							
A14	Enable operation & use	ITSM203 ITSM303							
A15	Procure IT resources	ITSM209							
A16	Manage changes	ITSM104 ITSM105 ITSM206 ITSM207 ITSM208							
A17	Install and accredit solutions and changes								
Deliver and Support									
DS01	Define and manage service levels	ITSM201 ITSM202 ITSM402 ITSM501							
DS02	Manage third-party services	ITSM201 ITSM202 ITSM210 ITSM402 ITSM502							
DS03	Manage performance and capacity	ITSM101 ITSM102 ITSM103 ITSM104 ITSM201 ITSM402 ITSM403 ITSM405							
DS04	Ensure continuous service	ITSM101 ITSM103 ITSM201 ITSM402							
DS05	Ensure systems security	ITSM101 ITSM102 ITSM103 ITSM104 ITSM105 ITSM208							
DS06	Identify and allocate costs	ITSM205 ITSM301 ITSM302 ITSM307							

The next stage is to map the COBIT control objectives to the IT goals. As you can see in the example below, there could be numerous IT underpinning objectives that support a single goal. Typically, the overarching IT strategy has only three to five high-level goals,

2 Example IT Goals	Identifier	IT Underpinning Objectives	Control Objective 1																Effectiveness	Efficiency	Confidentiality	Integrity	Availability	Compliance	Reliability
			P09	A12	ME02	AC3																			
Reduce Cost & Complexity	1	Execute risk based software compliance approach	P09	A12	ME02	AC3																			
	2	Execute tools simplification agenda to reduce cost & complexity	P05	DS12	DS13																				
	3	Reduce solution and service delivery defects / rework	P08	DS10																					
	4	Ensure expectations are set and services aligned with business requirements	P06	DS01	DS02	DS03	DS04	DS05																	
	5	Ensure IT Services remain competitively priced	P05	DS03	DS06	ME01																			
	6	Manage IT-related business risk	P09	DS05	ME02	ME03	ME04	ME05	AC01	AC03	AC04	AC06													
Improve Execution	7	Increase ability to absorb rapid change with acceptable risk	P09	A16	A17	DS09	AC1	AC3																	
	8	Balance IT View (Stability vs. Responsiveness, Reactive vs. Proactive, Quality vs. Cost)	P01	P05	A16	DS01	DS13																		
	9	Ensure IT services & infrastructure can properly resist & recover from failures due to error or disaster	P09	A12	A13	A16	DS04	AC3																	
	10	Create a strong IT foundation	P01	P02	P03	P04	P05	P06	P07	P08	P09	P10													
	11	Optimize the IT infrastructure, resources and capabilities	P04	P08	DS01	DS03	ME01																		
	12	Increase responsiveness to changing business requirements	P04	P09	P10	A16																			
Resource Optimization	13	Accurate prioritization of business requests	P04	P09	P10	A16																			
	14	Ensure proper use and performance of the applications and technology solutions	DS03	DS07	ME01	ME02	ME03	ME05																	
	15	Leverage Control Objectives, ITSM process entrance / exit criteria & SDLC Stage Gates for alignment	P04	P09	P10	A16	A17	ME05																	
	16	Leverage ITSM to accurately represent total cost of IT services	P05	DS06																					
	17	Ensure transparency & understanding of IT cost, benefits, strategy, policies	P01	P04	P05	DS06																			

so it is typically a one-to-many relationship with underpinning objectives.

The final stage is to map the IT goals/objectives to the business goals. The example below surfaces areas where IT can focus their efforts to have the most impact on achieving the business goals. Now, you can trace these back to the original capability

Perspective	Identifier	Business Goals 1	IT Goals / Objectives 2																Effectiveness	Efficiency	Confidentiality	Integrity	Availability	Compliance	Reliability
			4	18																					
Operational Perspective	1	Align IT Service Quality (Fit for Purpose / Fit for Use)	4	18																					
	2	Increase IT Resource Productivity	2	3	8	11	14																		
	3	Increase Operational Capabilities	3	10	11	13	14																		
	4	Manage Information Confidentiality, Integrity, and Access	1	6	9																				
	5	Increasing the agility of IT Service Offerings	7	8	12	13																			
Financial Perspective	6	Ensuring a good ROI on IT-enabled business investments	3	11	14																				
	7	Leveraging IT to manage technology related business risk	1	6	7	8	9																		
	8	Maintaining IT service cost alignment with open market	2	3	5	8	11																		
	9	Reduced contract costs and holding suppliers accountable	1	2	16	17																			
	10	IT Cost Control via Resource Optimization	2	3	5	11	15	16	17																
Strategy Perspective	11	Improve position of IT services for competitive advantage	2	3	5	11	12	17																	
	12	Position IT to support business agility requirements	7	12	13	17																			
	13	Increase Customer Retention via IT Service Experience	3	8	12	14																			
	14	Maintain alignment with Vision, Mission, Core Values	7	8	9	10	11	12	17																
	15	Transition IT into an "Integrated Business Partner"	4	5	6	7	8	10	12	15	17														
Industry Perspective	16	Increase Market Share & Position with IT Advantage	5	6	7	9	10	11	12																
	17	Balanced Investments to drive acquisition & retention	8	11	15	16	17																		
	18	Leverage technology for broad distribution	4	7	9	10																			
	19	Leverage technology for brand leadership	5	10	17																				
	20	Acquire & maintain skilled and motivated people	10	11																					

A documented journey and roadmap can be very powerful communication tools and help prevent your ITSM journey from stalling.

sheet so that you can see potential focus areas for IT investments.

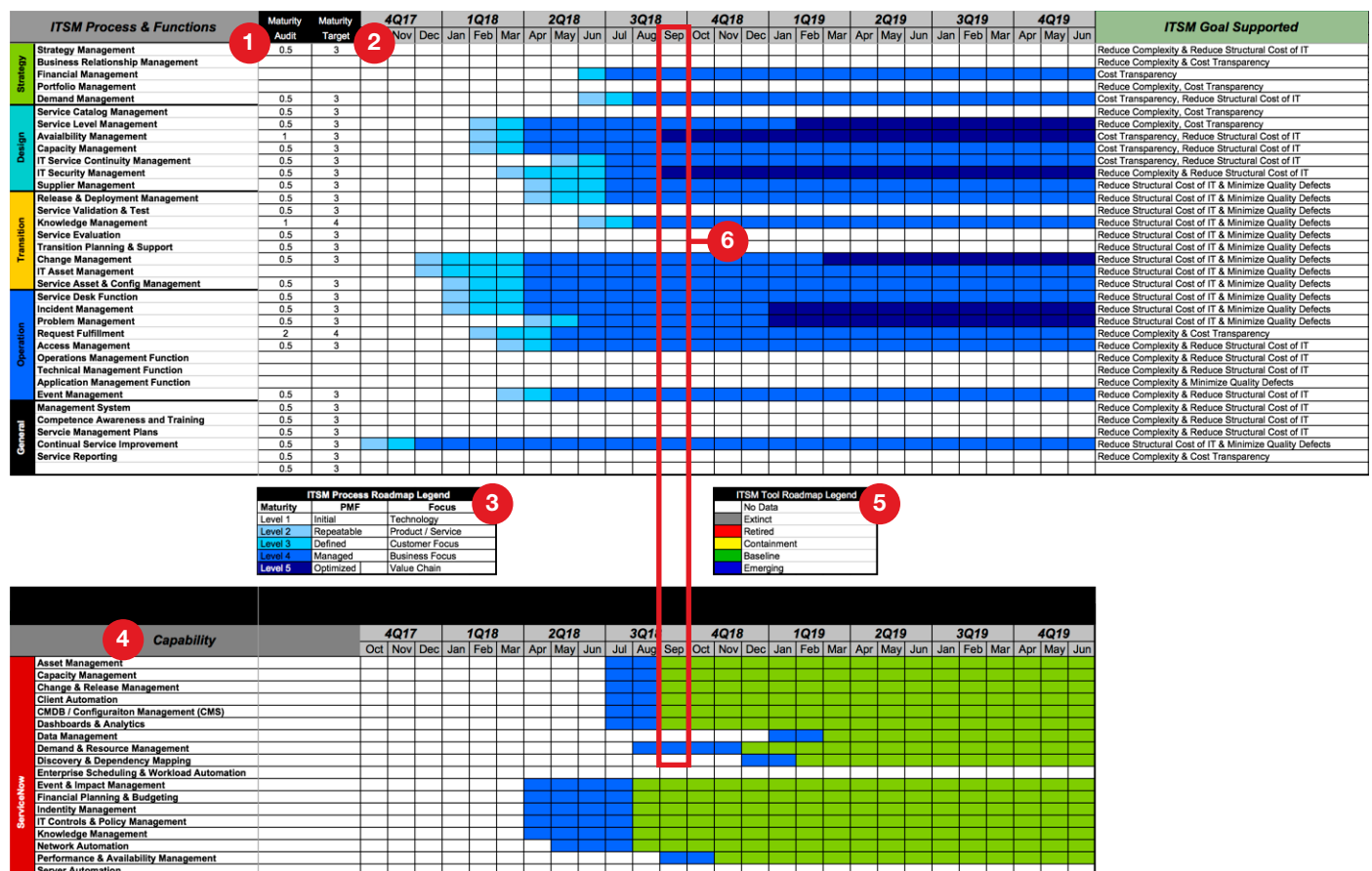
Activity 2: Documenting the journey

The capability mapping exercise performed earlier will serve as the primary input for completing this journey activation map. Creating a series of stages similar to those below, along with color codes, helps illuminate the organization's activation journey. When transitioning from legacy ITSM, the goal should be to achieve capability parity as quickly as possible without creating unnecessary risk from an operational perspective.

A documented journey and roadmap can be very powerful communication tools and help prevent your ITSM journey from stalling. The diagram below serves as the basis of mapping out your activation journey. The recommendation is to color code the journey stages and specific capabilities planned for each stage of the activation.

The final step is to transition the results of the mapping exercise into the ITSM roadmap. This collateral is color coded to illuminate what a journey may look like for a typical customer, but each journey will be unique to the specific organization. This serves as powerful communication tool but it also helps maintain alignment with new capabilities being delivered by ServiceNow.

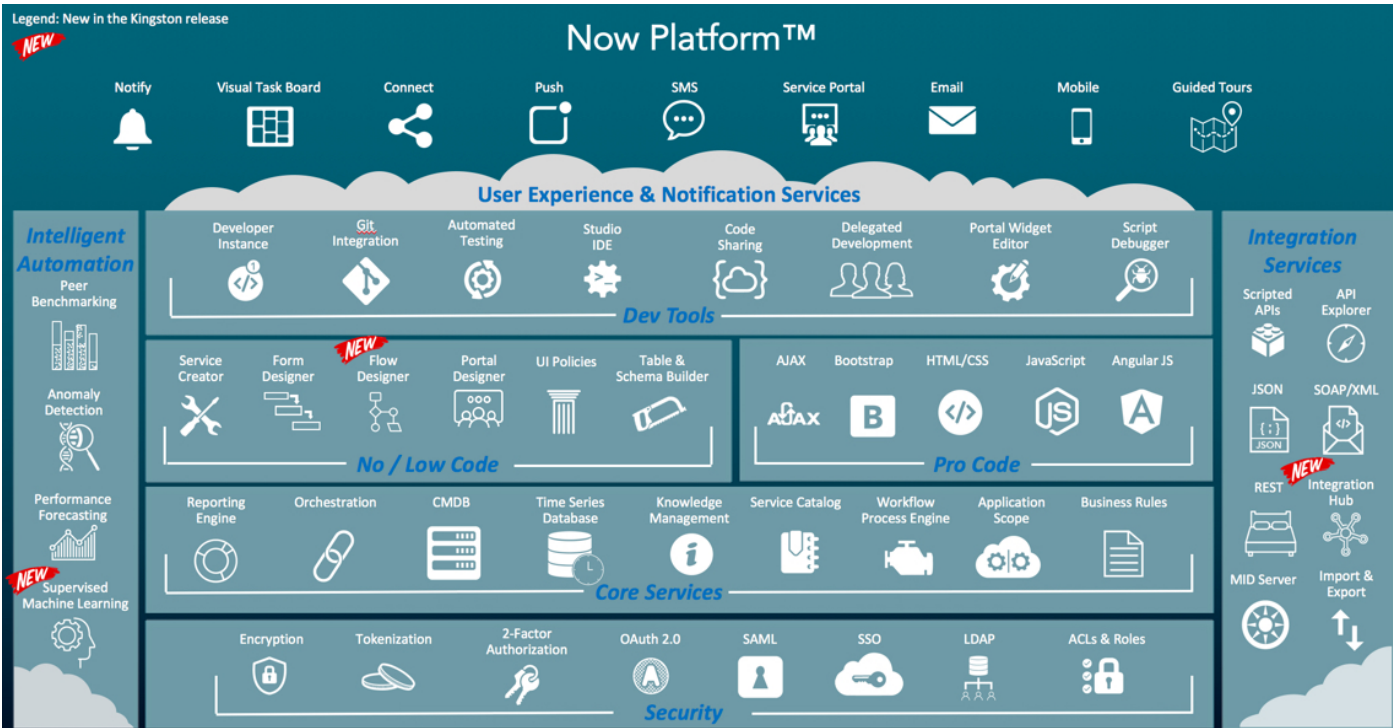
In the example below, callout #1 and callout #2 show the results from the last maturity assessment and the target maturity respectively. Callout #3 and callout #5 represent the color-coded legends. Callout #4 represents the underlying technology component, and callout #6 shows the top to bottom alignment of process and technology.



Platform and architecture

The Now Platform delivers an intelligent system of action for the enterprise. Using a single data model, it is easy to create contextual workflows and automate business processes. The platform's Intelligent Automation Engine combines machine learning with automated actions to dramatically reduce costs and speed time to resolution.

Anyone from business users to developers can easily build intuitive applications with modern experiences. With service catalogs to make requests, knowledge bases, and realtime notifications accessible to all users, you can deliver proactive and frictionless service. Departments, work groups, and even devices can assign, prioritize, collaborate,



Now Platform at a glance

and get down to root cause issues. Now, your business moves faster.

The Now Platform at a glance diagram is a good representation of the rich set of capabilities that underpin your ITSM instance. From skilled software engineers wanting to build scoped applications to process owners modifying application behaviors, the dev tool kit has a solution for you (pro code, no/low code). There are also core services that can be invoked from any OOTB or custom-built application, a comprehensive set of integration services, and the Intelligent Automation Engine all delivered via a scalable, highly secure cloud-based platform.

Most organizations have hybrid architectures that include cloud, on-premise technologies, and multi-provider service delivery models.

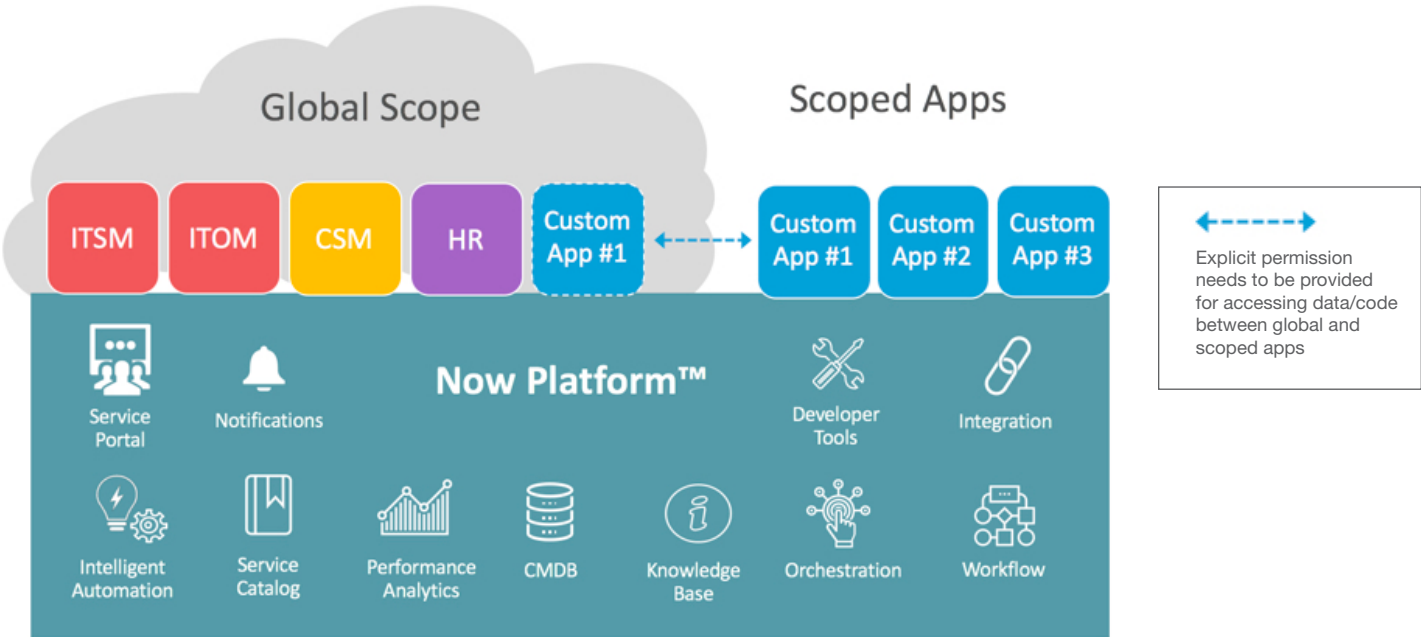
Architecture and integration services

ITSM ecosystems are more complex than ever before. Most organizations have hybrid architectures that include cloud, on-premise technologies, and multi-provider service delivery models. This makes architecture more important than ever before. It is critical that integrations be built using sound architectural principles and that they are well documented. TOGAF provides a rich set of proven guidelines when it comes to architecting your ITSM ecosystem. For more complex integrations, you will want to use skilled professionals who have a proven track record of developing integrations. If the organization does not have the skills in house, the ServiceNow partner network can be an excellent source of highly specialized skills.

The Now Platform offers a wide range of industry standard integration technologies (e.g., REST, SOAP, JDBC/ODBC, Java API, LDAP) so that you can integrate with nearly any system or subsystem on the market. These standards-based integration technologies allow customers to unleash the power of the ServiceNow single system of action. With these rich integration services, business users are able to interact with fewer disparate systems via the modern ServiceNow user experience (UX). In addition, there is a native capability for importing and coalescing structured data. The integration architecture provides a wide range of options so that the integration design can be aligned with existing skills and technologies.

Application scoping

The ability to create scoped applications on the Now Platform is an incredibly powerful capability that should be factored into your strategy. It is also a key architectural framework consideration as you build new or extend OOTB capabilities. Let's begin by examining the base characteristics of applications in the global scope



Applications in the global scope are like shared resources that any application developer can modify.



and scoped applications.

Global scope

The global scope is a special application scope that identifies applications developed prior to application scoping or applications intended to be accessible to all other global applications. Applications in the global scope do not append a unique namespace identifier to the application name. Global applications can have naming conflicts and data collisions when developers create multiple global applications with the same name.

Applications in the global scope are shared resources that any application developer can modify. Since global scope applications do not have a unique namespace identifier included in their application artifact names, they can have their own application access permissions. Global applications allow other global applications access to their tables to (1) read records, (2) run API requests, and (3) create configuration records.

Typically, only applications provided by ServiceNow are in the global scope. However, any custom applications created before application scope was implemented may also be in the

Important note: Applications in the global scope are not eligible for upload to the application repository or the ServiceNow Store. Since all global applications are in the same scope, they bypass scope protections. It is not recommended that any application in the global space be configured or customized to take on a lot of technical debt as it can create unnecessary upgrade complexities.

global scope.

Scoped applications

The most common use case for scoped applications is one where the customer wants to extend functionality to support another area of the business. With the help of application access settings, you can limit what is available to other parts of ServiceNow from your application. As an example, a conference room booking application can have its own tables and business logic in the conference room booking private application scope. While it can allow other applications to read its records, such as a list of available conference rooms, it can prevent other applications from overwriting protected data such as reservation schedules.

Plugins

Plugins are software components that provide specific features and enhanced functionalities within a ServiceNow instance. New plugins become available with each new release. Certain plugins are activated by default on every new instance. When you upgrade, most plugins aren't enabled on your upgraded instance. If you want additional plugins, you must activate them. Users with the admin role can activate some plugins through self-service. Other plugins must be requested through HI and activated by



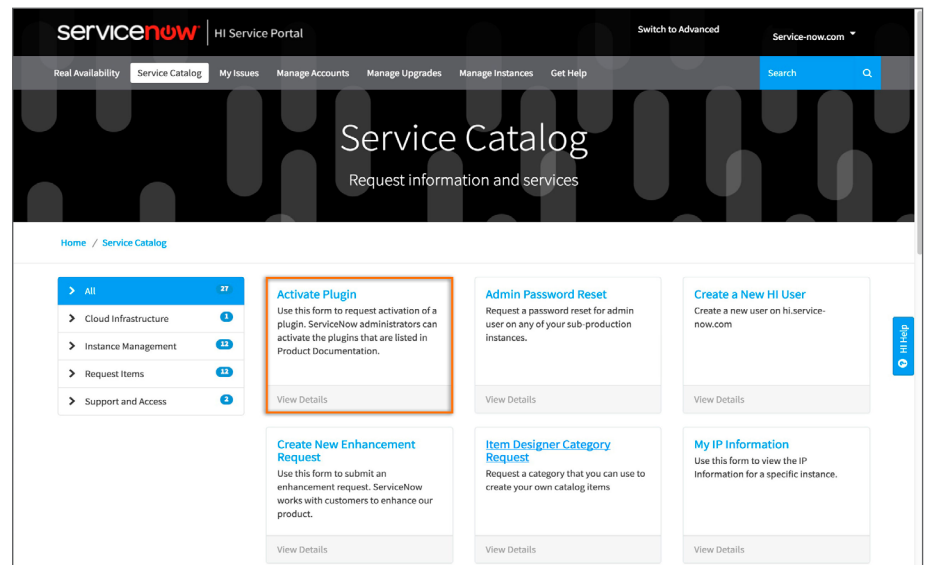
Important note: Plugins cannot be deactivated once they are activated, with a few exceptions. Customers should thoroughly review and test plugins in a sub-production instance. If they don't like the plugin, they can clone or zBoot (zBoot = full reset) the instance to clean it out. You cannot do that in production.

There are approximately 1,500 plugins available in ServiceNow. Some of these are subscription based but many are free.

ServiceNow personnel, either because they require a subscription or for some other reason—for example, because they are appropriate only for certain deployments.

When activating a plugin on your ServiceNow instance, follow these best practices.

- Avoid mistaking one plugin for another.
- Write down the name of the plugin you want. Some plugins have similar names that can be mistaken for each other.
- Check to see if the plugin is already active on your instance.



If the desired plugin isn't listed in the **System Definition > Plugins** module, request the plugin through HI by navigating to **Service Catalog > Activate Plugin**.

There are approximately 1,500 plugins available in ServiceNow. Some of these are subscription based but many are free. If there are plugin dependencies, they will be loaded automatically. When submitting a HI request for plugins, it is important to note the correct plugin name and submit only one activation and one instance on each request. This is done to comply with our change management policy.

Foundation for success

One of the cornerstones of any ITSM program should be the ongoing emphasis on building and maintaining a strong foundation. This begins with ensuring that there are always clear lines of authority and accountability. The following six activities are recommended when it comes to building and maintaining a strong foundation.

Activity 1: Creating an activation plan

Once the ITSM strategy, roadmap and journey are mapped out, developing a “100-day plan” is recommended. A strategy document can feel a bit overwhelming to the team, so developing a 100-day plan brings it down to something consumable, tangible,

It is important to vet and agree on a common lexicon for your entire services organization so that everyone shares a common language and understanding early in your journey.

actionable, and achievable for everyone. It also serves as a mechanism to secure early wins and build team confidence.

The goals in the 100-day plan should be short in duration and have a very specific objective. This may seem very tactical in the context of strategic planning, but it is frequently used to chart out the initial execution steps of the digital transformation journey. Strategy and vision are typically long term (about two to three years) perspectives so having an initial plan to activate the team can be very useful. In some cases, the 100-day plan may include actions such as:

- Service portfolio reviews and fleshing out consumer models
- Identifying overlapping work streams and inefficiencies
- Business user surveys and focus groups
- Completing the operating model RACI
- Creating a common lexicon
- Defining roadmap and success criteria

Activity 2: Establish a common lexicon

It is important to vet and agree on a common lexicon for your entire services organization so that everyone shares a common language and understanding early in your journey. This includes cross functional representation from service management, enterprise architecture, security operations, business operations, and others. Bringing in terms and definitions from the various frameworks often serves as a basis for your organization to land on a common definition.

Most of the framework methodologies offer up their own lexicon but unfortunately, there is little or no alignment among them. Therefore, common definitions may need to be modified for your organization. While having a discussion around “What is a business service?” may seem painful, it can be incredibly value when it comes to getting everyone

Term	Definition
Service	A service delivers value for customers. It has thre aspects: the interaction, the offering, and the service system.
Capability	An ability that an organization, person, or system possesses. Capabilities are typically expressed in general, high-level terms and typically require a combination of organization, people processes, and technology to achieve.
Feature	A feature is a distinguishing characteristic of an organization, person, process, or technology. (e.g., performance, portability, or functionality).
Process	A set of activities designed to accomplish a specific objective or goal. A process takes on one or more defined inputs and turns them into defined outputs.
Function	A function is a collection of people, tools, or other resources used to carry out one or more processes or activities—a collection of organized capabilities.
Configuration item	Configuration items (CIs) are physical and logical components of an infrastructure that are currently, or soon will be, under configuration management. CIs may be a single module, such as a server, operating system, application, and a physical or logical service, that make up a complete system.

aligned. It also serves as mechanism to bring the various business units together early on in the journey. This can be a great first assignment for your newly formed ITSM-ARB.

Activity 3: Publish an operating model RACI

Creating an operating model RACI is a high-value, low-complexity exercise that is an essential building block when it comes to creating a strong foundation. Notice the example below includes business roles, external service providers, and the actual output associated with each activity.

The recommendation is to start with the ITIL v3 2011 processes, but tailor those to your organization and include any scoped application areas that may have process elements. ServiceNow also recommends that you add a “Deliverable” field for every operating model activity. If there is no concrete deliverable or intended outcome, then it may be an activity that should be re-examined in terms of usefulness. If you are in a SIAM

ServiceNow Customer Operating Model RACI v1																								
Process Activities		Business	CTO Office	Relationship & Portfolio Mgmt					ITSM Process Owners							Managed Service Provider								
1 Activity	2	Business Sponsor	Business Customer	Enterprise Architecture	Digital Strategy & Innovation	Head of IT Strategy	IT Business Analyst	Portfolio Manager	Project Manager (EPMO)	IT Change Advisory Board	IT Release Readiness Review Board	IT Change Manager	IT Configuration Manager	IT Release Manager	Service Catalog Management	IT Infrastructure Manager	End User Computing	Solution Architecture	ICC Manager	Supplier Engineering	Supplier ITSM Organization	Supplier Architect	3	4 Deliverable
ITSM Modernization																								
Service Operation (Service Desk Function)																								
Detection & Recording		R										R		R				R		R				New / Updated Incident Record in ITSM
Classification & Support																		R		R				Properly categorized Incident in ITSM
Investigation & Diagnosis (Initial)																		R		R				Initial investigation completed and re-categorized or re-prioritized as needed
Incident Matching																		R		R				Complete Incident Matching query against all incident repositories
Check Knowledge Sources																		R		R				Completed check of all Knowledge sources (Internal & External)
Check Known Error Database (KEDB)																		R		R				Completed check of Known Error Database (KEDB)
Attempt 1st Call Resolution																		R		R				Completed attempt at First Call Resolution (FCR)
Resolution & Recovery	C	C																R		R				Resolved Incident
Incident Closure	I	I																I		I				Closed Incident
Distribution of IT Communique	I	I																I		I				Incident closure notification via email
Service Operation (Incident Management)																								
Verify Impact & Urgency		C											R	R				R		R				Verified impact & urgency along with updated incident record if appropriate
Check all Knowledge Sources													R	R				R		R				Completed check of all knowledge sources (Internal & External)

environment, this model should include external service providers as well.

Activity 4: Define ITSM scope & context

Establishing program scope is important so that there is clarity when it comes to authority and accountability. Scope often incorporates the entire ITSM ecosystem, which includes internal and external service provider touch points. It is recommended that organizations be very intentional when designing the program and processes so that they are extensible enough to account for enterprise and service delivery models regardless of who supports it, where it is sourced from, or what service is being provided. Maintaining discipline helps minimize the risk of scope creep, drift, and the

As early as possible, validate and recalibrate the consumer model of the various services, relationships, utilities, and assets along with the specific customer outcomes they fulfill.

temptation to over engineer a solution.

Activity 5: Validate the consumer model

As early as possible, validate and recalibrate the consumer model of the various services, relationships, utilities, and assets along with the specific customer outcomes they fulfill. These models can be used by the various process design teams, they establish a basis for the Service Catalog, and they greatly accelerate any software asset and configuration management (SACM) initiatives. Once the initial service model

Sample: Business discovery/validation questionnaire

✓ Who are the consumers?

✓ Who are the providers?

✓ How does this service create value?

✓ What is the value?

✓ Is this a strategic or commodity service?

✓ How does pricing compare to the open market?

✓ How does cost compare to our competitors?

✓ What changes might increase the value of the service?

✓ How do consumers want to engage the service?

is constructed, the service manager can prepare a business discovery/validation questionnaire.

Additionally, this modeling activity should include making clear distinctions between the different service provider archetypes such as internal service providers (Type I), shared services units (Type II), and external service provider types (Type III) to maximize the clarity of the offerings (see below). This initial work takes time, but it has enormous value

Type I (internal service provider)	Type II (shared services unit)	Type III (external service provider)
• Tend to operate within internal market spaces • Typically, business functions embedded within the service units they serve • Typically funded by overhead and tend to operate within the mandates of the business	• Focus on business units as direct customers • Create, grow and sustain internal markets for their services • Model themselves alongside service providers in open market • Are vulnerable to open market service providers	• Can offer competitive pricing and drive down costs by consolidating demand • Increased flexibility and freedom to pursue opportunities • Economies of scale • Expertise not limited to any one enterprise or market • May have specialized skills and expertise

Invest in a formal and ongoing ITSM awareness campaign.

in terms of meeting objectives and feeding the service management program initiatives, which are ultimately anchored in delivering value in the form of business outcomes. Please note a given service provider can have more than one type.

Activity 6: Organizational change management

Cultural changes can be complex, and often, the mere mention of a “technology refresh,” “process,” or “framework” will activate the organization’s autoimmune system. However, messaging around value delivery in the form of business outcomes and re-positioning some of the “processes” and “functions” as “service offerings” can overcome much of the cultural resistance. For example, getting process owners and consumers to start thinking in terms of a portfolio of change or release *services* versus a change or release management “process” will help keep the emphasis on serving rather than control. It is important to use messaging that is anchored in value propositions and tailored to the audiences.

Invest in a formal and ongoing ITSM awareness campaign. Treat the modernization initiative as if a new product or service were being launched to the external market. Elevating the image of ITSM is part of transforming IT into an integrated business partner. It requires ongoing, clear, and consistent messaging that continually re-enforces the value propositions. It is important to seek out and recruit change agents, apply industry knowledge, and engage internal marketing teams to help develop a campaign that will resonate and ultimately, help institutionalize service management in the organization.

The example below is a business capability map that is closely aligned with the

MINDSET	MODERNIZE	TRANSFORM	INNOVATE
CONSUMER	IT Professionals	IT Customers	All Employees
PHASE	CONSOLIDATION	EXPERIENCE	ALIGN
CAPABILITIES	Incident Change Request Knowledge CMDB	Orchestration Problem Asset Release Analytics	Event Service Mapping Mobile Service Catalog
BUSINESS VALUE	Simplify & Standardize	Self-Service & Automate	Visualize & Enrich
OUTCOMES	IT SERVICE CONTROL - Standardized processes consolidated on common cloud platform - Inherent process integration and automation - Common request catalog for IT services - Knowledge articles to enable deflection - Populated and trustworthy configuration data aligned to incidents and changes	IT SERVICE ENGAGEMENT - IT Portal for Self-Service and Engagement - Extended problem reporting & root cause analysis - Mature asset inventory aligned to financial awareness - Automated and governed release process tied with agile methodology - Analytics dashboards collecting appropriate metrics and reporting on process	IT SERVICE ALIGNMENT - Enhanced CMDB with mapping of Services and Business Applications - Mobilized ITSM Experience for employees and IT Customers - Service Portfolio visualization and Service Owner awareness - Event Management and Alert integration to drive incident automation

Modernize IT service management

ServiceNow technology stack. It can be a great way to represent your activation journey in PowerPoint to stakeholders. This example represents a three-stage customer activation journey.

A well-executed service management awareness campaign will likely create an enormous energy swell, which will get everyone energized, but it is important that it extends beyond the initial transition period. As IT professionals, our comfort zone tends to revolve around the technology and *how* things are done over *why* things are done. However, it is important to communicate not just the *how* and what but also *why*. The

Service desk best practice focus areas

- Omnichannel
- Automation
- Use of templates
- Digitize/shift left
- Tech lounges

why illustrates, to the customers, the value story. *Teach the organization how and they will simply check the boxes. Teach the organization why or the value association and they will be inspired.*

ITSM recommended best practices**Service desk**

The service desk is often perceived as the face of IT, so your ITSM strategy should clearly articulate both the customer experience expectations and the desired business outcomes. While the service desk function is just one element of the overall customer experience, it plays an important role when it comes to the lifecycle management of incidents. A modern service desk can compensate for inefficiencies in other areas of the IT ecosystem by delivering exceptional customer service, holding service providers accountable to their contractual obligations (e.g., SLAs, OLAs, UCs), and distributing customer-friendly status updates and standardized business communications.

Service desk extensions such as customer-facing self-service portals, virtual agents, tech lounges and the new ServiceNow walk-up experience can play an important role in customer satisfaction. Bringing services to business users in ways that meet their consumption preferences dramatically improves the user experience. It is a misconception to think of the service desk as a “ticketing system.” The perception of IT is often determined by the respective business user experience(s).

Let’s examine some of the best practice recommendations for building a modern service desk using ServiceNow. They include (1) Omnichannel support, (2) the use of machine learning, virtual agents and automation, (3) use of templates, (4) continually digitizing and shifting left, and (5) offering an integrated walkup experience (e.g., Tech cafes).

(1) Omnichannel support

A key activity of the digital enterprise journey includes identifying techniques that make information easily accessible and brings solutions closer to the business users. This includes embracing their consumption preferences (e.g., Self service, tech lounge, virtual agent, live agent). Regardless of the channel, it is important to capture all interaction details. This ensures that information can be reused by both humans and machines. When agents capture good detail and generally normalize data captured in an incident record, that information can be accessed and reused by other agents (e.g., incident matching), evaluated as potential knowledge, and serve as a rich information source when it comes to building prediction models for machine learning.

(2) Machine learning, virtual agents, and automation

These emerging capabilities are fundamentally changing the way we do business. These capabilities are visible in the consumer markets today but corporate IT is just beginning to realize the full potential of these disruptive technologies. It is misconception to think of these as human replacements. The real power and value comes from understanding how to augment the IT workforce.

Virtual agents can perform routine actions quickly and surface potential resolution options to business users as they type (e.g., contextual search). It is recommended that customers adopt a hybrid approach when it comes to chatbots that includes natural language processing (NLP), natural language understanding (NLU), conversational interface (CI) and machine learning to produce conversation. A chatbot perspective

The goal of incident management is to restore normal service operations as quickly as possible after an unplanned interruption or service degradation.

should be incorporated into your overarching ITSM strategy.

(3) Use of templates

Templates usually have predefined steps to handle issues that are seen repeatedly and require the same steps to resolve each time. Examining the incidents coming into the service desk will likely provide some insights into patterns and incidents that hit the desk routinely. Instead of agents filling out the same incident form over and over again, they can build a template to quickly log common incidents and autofill required fields including the assignment field, if necessary. This ultimately reduces the handling time, eliminates that potential for human error and normalizes the record content.

(4) Continually digitize & shift left

Providing easy access to key intelligence, moving solutions closer the business, using BOTs, and machine learning have all become key elements of shift left initiatives in the digital enterprise. While modernization and the digital enterprise have not fundamentally changed the underlying principles of ITIL, they have completely changed that way that we expose knowledge and engage the business. At ServiceNow, our intent is to give business and IT users the consumer-like experience that they enjoy in their personal lives but with a richness that is unparalleled in the industry.

(5) Tech lounges and walk-up experience

Tech lounges, tech cafes, and the ServiceNow Walk-up Experience are a huge value add for most business users. They tend to be easy to set up and have a low cost footprint. With the new ServiceNow walk-up experience, we make this a native capability for our customers. This is an area that analysts predict will continue to expand in the coming years and it has a proven track record for success when it comes to customer satisfaction.

Incident best practice focus areas

- Categorization Management
- KPIs and benchmarks
- Major incident management
- Discipline
- Single system of record

Incident management

The goal of incident management is to restore normal service operations as quickly as possible after an unplanned interruption or service degradation. The practice of incident management is well understood across the industry but there are often additional best practices that can be applied to fully optimize the incident management service.

Let's examine some of the recommended best practices to improve the overall efficiency and effectiveness of Incident Management using ServiceNow. Recommended best practices for modernizing your Incident Management program include (1) periodic categorization reviews, (2) using KPIs and ServiceNow Benchmarks, (3) maintaining a structured program around Major Incident Management, (4) maintaining incident management discipline, and (5) leveraging the ServiceNow single system of record.

(1) Categorization reviews

Categorizations tend to drift over time as the business needs change. This can present challenges for agents when it comes to incident categorization and it increases the likelihood of incorrect assignments. Grooming the categorization schema regularly is recommended to ensure that your categories are up to date, and the schema is optimized for the organization.

A good rule of thumb is to keep it as simple as possible for your organization. For example, a smaller organization may only want a dozen categories, whereas a larger and more complex organization may want a multi-level hierarchy. Always keep simplicity in mind since the more elaborate the categorization schema is, the more complexity it

ServiceNow Benchmarks are unique to the industry and serve as an incredibly powerful tool for continual improvement.

introduces into your system.

(2) KPIs and ServiceNow Benchmarks

ServiceNow Benchmarks are unique to the industry and serve as an incredibly powerful tool for continual improvement. The ability to compare KPIs with like organizations, in the same geography or in the same market segment can be very helpful. Benchmarks provide a visual trend graph so that you can monitor trends over time and avoid reacting to anomalies. Benchmarks also include a set of ranked recommendations for KPIs that are underperforming when compared to the global benchmark. This is an incredibly powerful capability that successful ITSM programs tend to use heavily.

(3) Program discipline

It is important that incident managers keep their focus on restoring normal service operations. In many cases, incident managers find themselves acting as both incident and problem managers. These are two very different disciplines that should have some degree of process uniqueness. Incident managers should be highly focused on doing whatever is needed to get business users operational again, while problem managers should be using a more methodical approach to solve for the underlying root causes of issues that have the highest cost to the organization. Keeping these two disciplines completely separate is not always possible but it is important that everyone understands what role they are playing at any given time.

(4) Major Incident management

One of the keys to successful major incident management is early detection. Early detection of a Major Incident (MI) can potentially avoid a widespread outage or significantly reduce the financial and business impact associated with the major incident. Any incident can be flagged as a major incident candidate and immediately reviewed for potential impact. The MI manager can monitor all candidates and promote them as needed. ServiceNow major incident management provides this early detection capability along with a major incident management single pane of glass for the handling all potential, open, and closed MIs. The MI workbench also includes a robust set of capabilities for business/technical communication, SME engagement, and outage management.

(5) Single system of record

The parent-child record relationship of records in the platform enables organizations to articulate a variety of relationships (incidents to incidents, incidents to problems, problems to changes, etc.). This is an incredibly powerful capability that is often underutilized in most organizations. With ServiceNow, this is native to the platform so our recommendation is to embrace this capability and build into your procedures.

Change best practice focus areas

- Balanced approach to risk
- Use of standard changes
- Emergency changes
- CAB meetings
- Conflict detection

Change management best practices

Most organizations have a reasonably well-defined change management process. However, there are typically big opportunities when it comes to optimizing change management. A change management program should serve as an enablement and risk assessment capability for the organization rather than just a control mechanism. Change management is sometimes viewed as an obstacle that gets in the way of innovation, but with a little tuning, change management can serve as an accelerator for delivering high-

Standard changes are one of the most underutilized optimization opportunities in ITSM.

velocity change in an organized manner. One of the tuning considerations is to secure the right mix of change types (standard, normal, and emergency) for your organization.

Let's examine some of the common practices that improve the overall efficiency and effectiveness of your change management program using ServiceNow. The recommended best practices for optimizing your change management program include: (1) having a balanced approach to risk, (2) developing KPIs to increase percentage of standard changes, (3) minimizing the number of emergency changes and (4) using the ServiceNow Change Advisory Board (CAB) workbench for highly productive CAB meetings.

(1) A balanced approach to risk

It is recommended that you take a balanced approach to risk evaluation. This balanced approach involves both a quantitative risk calculation combined with a qualitative risk assessment. Risk evaluation helps identify factors that may disrupt the business or impede the delivery of services. Different approaches can be taken to assess and manage risks, and each organization must determine the approach it wants to use. Different ways to determine risk include the following:

- Conflict detection (forward schedule of change)
- Risk assessments (questionnaire)
- Risk conditions (calculated)

(2) Standard changes

Standard changes are one of the most underutilized optimization opportunities in ITSM. Standard changes are preapproved, low-risk changes with repeatable implementation steps and a proven history of success. For these frequently implemented changes, time and effort can be saved by creating standard changes. These changes are pre-authorized by Change Management or the CAB. The base incentive for teams to use standard changes includes the ability to go fast in a predictable manner without the typical overhead associated with the normal change workflow. This is a very attractive option for DevOps teams.

Thinking through the change ahead of time and carefully documenting all the steps, release windows, and the associated risk allows change implementers to submit these recurring, low-risk changes to the CAB for standard change consideration. Once the CAB approves it as a standard change, teams are able to move freely within the agreed upon conditions of the standard change. In the event that the standard change fails or results in change-related incidents, it should go back to CAB for reconsideration.

Building out the standard change library is important and should be accompanied by an awareness campaign. It is also recommended that you set specific KPI targets to drive standard change adoption. A typical KPI goal might be to increase the number of standard changes by 20% in the next quarter. This allows the team to focus on driving to this target. The recommended percentage of standard changes is the 70%–90% range.

(3) Emergency changes

Emergency changes inherently carry more risk than normal or standard changes but they are an inevitable part of change, and therefore it is recommended that you have a specific policy around the handling of emergency changes. Some organizations allow for latent changes, expedited changes, and fail forward plans. In general, these variations are not a recommended practice but there are situations when they may make sense for a period of time. Even emergency changes should maintain some degree of formal rigor and reporting. The absence of a healthy standard change library can put an unnecessary load on the normal change process which may result in a higher ratio of

ServiceNow CAB workbench is a modern approach to running a CAB and it will change the way you view CAB meetings forever!

Problem best practice focus areas

- Tiered RCA
- Proactive problem management
- Continual reprioritization
- Creating knowledge
- Application areas

emergency changes.

(4) CAB Workbench

If you have ever manually built a CAB agenda or attended a traditional CAB meeting, you probably found yourself sitting through a multi-hour meeting listening to change presentations while patiently waiting to represent your change. ServiceNow CAB Workbench is a modern approach to running a CAB and it will change the way you view CAB meetings forever! CAB Workbench includes a wide range of capabilities that allow for productive meetings where stakeholders are notified right before their change is presented so that they only have to attend the parts of the meeting that are relevant.

CAB Workbench also saves change managers a lot of time preparing CAB agendas, facilitates better stakeholder engagement, and maintains a clear record of related decisions. It includes an agenda with specific time slots for changes to be presented.

And most importantly, CAB Workbench benefits all organizations regardless of size, vertical, or geography.

Problem management

Problem management is commonly an area where there is a great opportunity to dramatically lower costs. The three primary objectives of problem management are (1) preventing problems and resulting incidents from happening, (2) eliminating recurring incidents, and (3) minimizing the impact of those incidents that cannot be prevented. Successful problem managers are typically savvy leaders equipped with a strong mix of analytical, technical, and business skills. They recognize the significance of making the right choices for both IT and the business.

Let's examine some recommended best practices to improve the overall efficiency and effectiveness of ServiceNow Problem Management. These include (1) a tiered approach Root Cause Analysis, (2) proactive Problem Management, (3) continual reprioritization, (4) creating Knowledge, and (5) applying Problem Management principles throughout the organization.

(1) Tiered root cause analysis (RCA)

One common barrier for problem management is that structured RCA often takes too long and doesn't always add value. When a problem team has to navigate through a series of complex problem solving techniques that may or may not add value, they will short-circuit this and eventually stop doing it entirely. The objective should be to rightsize the RCA model for the problem being solved.

There are cases when something as simple as the "5 Whys" is adequate for determining root cause and other cases where a more detailed RCA methodology has enormous value. The recommended practice is to attempt to solve for root causes using a timeboxed "5 Whys" RCA and then transition to something more structured such as Kepner-Tregoe or Ishikawa's Fishbone for those problems that require more detailed analysis. The RCA output can then be attached to the problem record as an attachment (e.g., Fishbone diagram).

(2) Proactive problem management

Service desk agents can play a key role in augmenting underfunded programs such as problem management. For example, assigning agents to search for recurring incident patterns during non-peak hours so that potential problems can be identified. Agents get

Problem managers should continually monitor relationship counts and reprioritize their backlog to ensure that they are always working on problems that have the highest ROI for the organization.

tired of solving the same incidents day after day, so this can be a very fulfilling assignment for them and it has a huge ROI potential for the organization.

It is recommended that problem managers periodically use a statistical technique such as a Pareto analysis (80/20 rule) to keep their focus on fixing the top 20% of problems which typically result in 80% of the benefit. Once that has been identified, problem managers can focus their attention on solving those problems that will have the highest impact and ROI to the organization.

(3) Continual reprioritization

Problem managers should continually monitor relationship counts and reprioritize their backlog to ensure that they are always working on problems that have the highest ROI for the organization. As a result, problem managers will be continually reprioritizing the problem backlog. Think about a scenario where a problem record is generated and it only has a dozen or so incidents related to it. It is something that has been identified as a problem but is not likely a top priority. During the reprioritization exercise a week later, the problem manager notes that the related incident count for that problem has jumped from a dozen to over 500. While the problem itself has not changed, the relative priority is very different.

(4) Creating knowledge

Knowledge is one of many potential sources of information for the service desk agents and business users. However, known errors do not always translate into the authoring of knowledge. Knowledge is often constructed in a more broadly consumable format and can be targeted to specific audiences. For example, Knowledge might be exposed via self-service to the community at large while a known error's information contained in a knowledge article may contain technical instructions (for example, a registry change) for a service desk agent to execute so that service may be restored quickly. Known errors are often more technically focused and can be very temporary in nature. The recommended practice is to create separate knowledge bases for information that should be restricted to certain audiences.

(5) Applying the principles everywhere

Problem management and the associated RCA methods can be applied to sub optimized processes, failed projects, and a series of other activities going on in the organization. Problem management principles can generally be applied anywhere and to anything.

CMDB best practice focus areas

- Scope definition
- Capturing requirements
- Rightsizing the program
- Automate discovery
- Service mapping

Configuration management database (CMDB)

It is recommended that only critical data be ported from CMDB to CMDB during the ITSM modernization initiative. It is often easier to rediscover CMDB data rather than attempt to port it from a legacy CMDB. This is also a great opportunity to re-evaluate the data model as well. It is recommended that only required data and attributes be imported into your new CMDB. This puts the SACM process owner in a position to extend the data as stakeholders require it. It is important to note that more data is not always better.

This is where it becomes critically important to factor the people pillar into your model by making sure the data can be properly managed. Fortunately, ServiceNow provides a wide range of automation, integration, and discovery capabilities, which can greatly reduce the human resources needed to manage what you may have been accustomed

Scoping decisions are often dictated by policies that address an organization's business drivers, contractual obligations, service commitments, governing laws, regulations, and standards.

to with your legacy tool sets. The CMDB is a repository of information about your IT infrastructure and services.

Let's examine some of the common practices that improve the overall efficiency and effectiveness of a CMDB using ServiceNow. The recommended best practices for optimizing your CMDB include (1) clear scope definition, (2) right sizing the program, (3) requirements management, (4) leverage service mapping and (5) being a key supplier for IT asset management (ITAM) and software license management (SWLM).

(1) Defining scope

Scoping decisions are often dictated by policies that address an organization's business

CMDB scoping considerations

- What environments will configuration management control?
- Which CIs in the CMDB need to be managed at the relationship level and which CIs require only inventory-or asset-level management?
- What IT services will be included?
- Are there geographic considerations?
- Are there regulatory or compliance requirements that must be met?
- Are there specific levels of control required for traceability and auditability?
- What security issues must be addressed?
- Are interfaces to internal and external service providers required?

drivers, contractual obligations, service commitments, governing laws, regulations, and standards.

Many of the answers to these questions may be stated as policies that will govern the design and development of the configuration management system, if they do not already exist.

(2) Rightsizing

One of the keys to a successful CMDB build is to rightsize the solution. Success hinges on identifying clear requirements and using field-tested best practices with respect to managing data. Having massive amounts of data with no ability to manage it will not produce good results. Our recommendation is to start small and expand.

(3) Capturing CMDB requirements

As the system of record for all information about the IT infrastructure, the CMDB is a foundational tool that supports the effective execution of all service management processes. Because of this, gathering requirements from all stakeholders prior to designing the CMDB is a critical first step for success. At a minimum, the following requirements must be determined:

- Governance requirements
- Inventory and asset requirements
- Regularity requirements

IT asset management (ITAM) and software license management (SWLM) are rarely overfunded or overemphasized in ITSM programs.

- Service management requirements

(4) Discovery and service mapping

Service mapping is an incredibly powerful capability that allows CMDB consumers to see the entire service delivery stack from top to bottom. The ability to visualize complex upstream and downstream relationships of CIs serves nearly every process area from incident, problem, change, release. For example, the ability to conduct business impact assessments when a change is proposed can avoid an unplanned outage.

ITAM and SWLM

IT asset management (ITAM) and software license management (SWLM) are rarely overfunded or overemphasized in ITSM programs. However, one audit or major technology refresh project will bring these capabilities to the value forefront. SWLM is critically important when it comes to protecting the organization from overbuying or underbuying licenses. Both of these conditions can be incredibly expensive and potentially have a very negative exposure for the company.

Conclusion

Thank you for choosing ServiceNow! We value every customer and partner. The content of this guide is intended to provide more perspective guidance on modernizing your ITSM practices on the ServiceNow technology stack for **(1) accelerated time to value, (2) dramatically lower costs, and (3) greater ROI**. Our hope is that you find this content helpful, and we encourage feedback as to areas that we should expand on in future releases of the publication. It should also be noted that specific application best

practices from a technology perspective can be found in the Customer Success Center and Champion Enablement at www.servicenow.com.

Glossary

A **service** delivers value for customers. It has three aspects: the interaction, the offering, and the service system.

A **configuration item (CI)** is a uniquely identified component used to deliver a service for which changes are controlled.

An **application** is specific software product installed on (specific) infrastructure used to perform a function.

A **zBoot** is a complete reset of your instance.

The **Information Technology Infrastructure Library (ITIL)** is a set of best practice guidance for IT service management (ITSM) that focuses on aligning IT services with the needs of business.

ISO/IEC 15504® is an excellent reference model for assessing process maturity. It is designed so that the assessors can assess the organization's capabilities for delivering products (software, systems, and IT services). It is also known as the Software Process Improvement and Capability Determination (SPICE) model and a derivative of the well-known Capability Maturity Model Integration (CMMI).

A **RACI matrix** or **linear responsibility chart (LRC)**, describes the participation by various roles in completing tasks or deliverables for a project or business process. It is especially useful in clarifying roles and responsibilities in cross-functional/departmental projects and processes. RACI is an acronym that was derived from the four key responsibilities most typically used: responsible, accountable, consulted, and informed.

ISO/IEC 20000 is an international standard for IT service management that includes "the design, transition, delivery, and improvement of services that fulfill service requirements and provide value for both the customer and the service provider."

Control Objectives for Information and Related Technology (COBIT), is a framework created by ISACA for information technology (IT) management and IT governance. It is a supporting tool set that allows managers to bridge the gap between control requirements, technical issues, and business risks.

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